Place-field repetition is what a predictive map must do when self-motion cannot resolve it: a cognitive map represents space only up to the symmetries self-motion cannot break

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Place cells fire at the same relative location in every identical compartment of a maze. This repetition is usually read as the hippocampus failing to tell the rooms apart. We show it is instead what a predictive code must do when its own sense of self-motion cannot resolve the symmetry, and that the same mechanism sets both the map and the network's capacity to replay.

A recurrent network trained only to predict its next observation (1) does not represent physical space X. It represents the quotient X/G: it writes every set of positions its self-motion signal cannot tell apart as a single place. The group G is not the environment's symmetry group. It is the subgroup *of* it that self-motion fails to break, fixed by equivariance and not by information, and both conditions are needed. Two head-direction encodings of one bit each, one invariant under a 180° rotation and one not, carry the same entropy yet separate completely in a symmetric arena ($p = 1.1 \times 10^{-5}$) and are identical where there is no symmetry ($p = 0.97$). Read through the C_4 quotient, the code falls to the group-theoretic $1/|G|$ ceiling.

What breaks the account is reconstruction, not prediction. A matched autoencoder discards the compass, folds every encoding alike ($p = 0.59$), and never generates more than $0.37\times$ waking coverage offline. One step of prediction recruits the compass, restores the fold, and lets the network replay. Weakening the compass then degrades the map and the replay together, in the same order. The two are therefore one variable, self-motion integration forced by prediction, read out in two ways, and this is what makes the account unifying rather than a fact about one measurement. A learned, unanchored compass builds the fold on its own.

The pure predictive code folds almost completely. A graded symmetry-breaking cue sweep, powered to resolve a genuine information ramp, found no rate-based signature of the partial repetition reported in rodents (2), only a small chance-exceeding residual already present with no added cue. A cognitive map represents space only up to the symmetries its own motion cannot break.

predictive learning | cognitive map | hippocampus | symmetry | place cells | head direction

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Introduction

In a maze of visually identical compartments, a place cell fires at the same location in each room: one cell reports several rooms as a single place. Spiers et al. (2) recorded CA1 place cells in four identical compartments arranged in a row off a connecting corridor, and found a high degree of spatial repetition. Grieves et al. (3) then ran the same four compartments in two arrangements. Kept in parallel, place cells often repeated their fields across compartments. Set 60° apart on a radial arm, so that a rotation rather than a translation relates them, the repetition largely disappeared. A separate cohort of rats also took significantly longer to learn to tell the compartments apart in the parallel arrangement than in the radial one, a behavioural cost that tracked the neural one (full description in Results). This repetition is usually read as a limit of the hippocampal map, an inability to tell identical rooms apart. We show the opposite. It is a necessary consequence of learning to predict, and it reveals what a learned spatial code represents when the world is symmetric.

Start with the simplest version of the problem. You are dropped into one of two identical rooms and asked which one you are in. Vision cannot help: the rooms look the same from every corresponding spot. Your only other source of information is your own movement, a sense of which way you face and how far you have walked. But sliding one room over to sit beside the other does not change which way you face inside it, and a compass reads the same in both. So no amount of tracking your own motion separates two rooms related by a translation: whatever you predict about what you will see next is identical in the two. Turn one room relative to the other, and your compass now *does* read differently between them, so your own motion can tell them apart.

That is the whole idea in miniature. What a self-motion signal can resolve is exactly the set of symmetries under which it is not itself invariant. A code learning only to predict its future observations must therefore collapse together every pair of positions its self-motion signal leaves looking identical, and keep apart only those it can distinguish. It represents space not as it is, but modulo the symmetries its own motion cannot feel.

Levenstein et al. (1) showed that a recurrent network trained to predict its own future observations develops spatially tuned units with no spatial supervision, and validated this in an L-shaped arena chosen so that every position has a unique visual fingerprint. Most rodent electrophysiology, however, uses square arenas, which are rotationally symmetric, and symmetric landmarks make distinct positions look identical. This is the classical perceptual-aliasing problem (4), and it raises a question the field has not asked of any learned spatial code: when distinct places look and feel the same, what does the map represent? The question is not peculiar to one model. The successor representation (5), the standard theory linking prediction to spatial coding, inherits it exactly, because positions related by a symmetry of the transition structure have identical expected future occupancy. Existing accounts largely assume the answer rather than derive it: the successor representation is defined over a state space that is supplied, continuous-attractor models (6) build the metric in by construction, and boundary-vector models (7) read firing from local geometry. None asks what a code *learned* from aliased observations must represent.

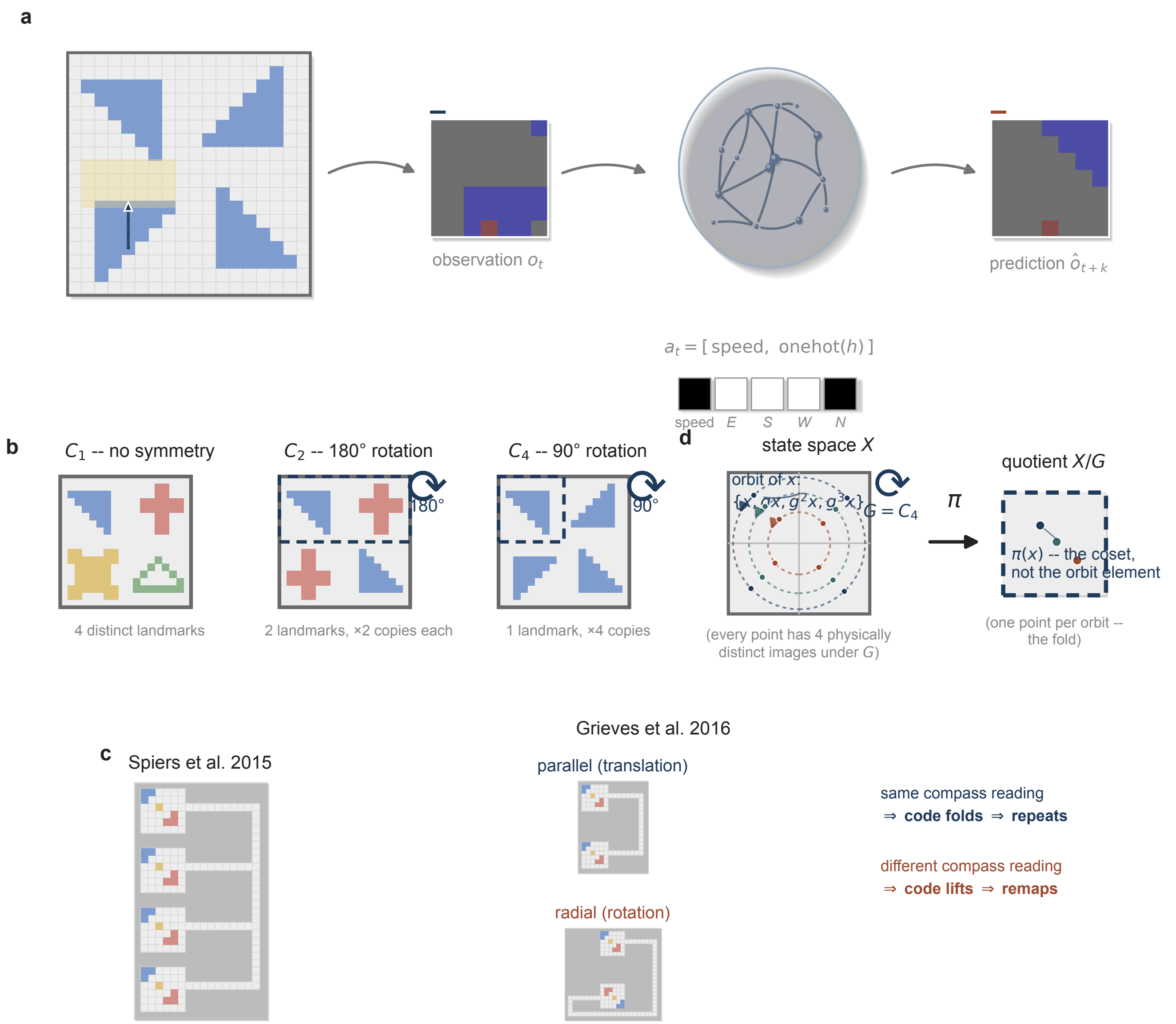
We show that the learned map represents not physical space X but its *quotient* X/G. It writes each orbit of symmetry-related positions as a single place, discarding which copy of the fundamental domain the agent occupies while keeping where it is inside that copy. In the language of Markov decision process theory this quotient is a *bisimulation*, the coarsest state abstraction that preserves a process's entire future (8, 9). Here it arises with no explicit abstraction step and no reward, both absent, simply as the state a network converges to when the only thing it is asked to do is predict. Two things fix the group G, and both are needed. First, G is a symmetry of the environment: with none, there is nothing to fold. Second, G is a symmetry the self-motion signal fails to break: a compass not invariant under a rotation resolves it, and the fold lifts. G is the intersection of the two, the symmetries the world has and the compass cannot feel, and every experiment here varies one factor at a time across it. Formally, if the head-direction encoding ϕ is invariant under a symmetry g of the arena, $\phi(h+g) = \phi(h)$, then the whole (observation, action) input stream is G-equivariant, the posterior over position is G-invariant, and no downstream reading of that stream can lift the ambiguity. What matters is therefore not the *amount* of information the compass carries but its *equivariance*, how it transforms under the group: an encoding carrying exactly the same number of bits, but not invariant under g, lifts the fold completely. Head direction is here an *instrument*, not the subject of the paper; it is the self-motion signal we can manipulate cleanly. The statement is general. Any self-motion signal resolves exactly the symmetries under which it is not invariant, and a pure speed cue, invariant under all of them, resolves none. Both halves are load-bearing. An invariant compass in an arena with no symmetry folds nothing: a network given no compass at all still separates x from its $180°$ image in the C_1 arena, at 0.966. And a symmetric arena read by a compass that breaks the symmetry folds nothing either.

The rest of the paper establishes this quotient law and its consequences. The fold shows first in single-cell tuning: fields grown with no spatial supervision repeat at symmetry-related locations when the compass is invariant, once under C_2 and four times under C_4. But it cannot be *read* from tuning alone, and much of what follows is why. An invariant compass makes the code look symmetric whether or not the arena is, so every single-cell measure of symmetry we tried reports a fold even in the C_1 arena, where none is possible; we therefore measure the fold throughout by orbit-phase decoding, the one readout whose null survives that arena. A controlled experiment that holds information fixed and varies only equivariance separates the two completely. Read through the C_4 quotient, the fold hits a parameter-free $1/|G|$ ceiling. It is built only under a *predictive* objective, since a matched reconstruction objective folds every encoding alike, and it arises unaided from a learned, unanchored compass, surviving one corrupted on a third of its steps. The law then earns its name against data: it explains why place fields in identical compartments repeat when the compartments are translationally related and separate when they are rotationally related (2, 3), and why abolishing the head-direction signal makes rotation-related rooms fold too (10). Throughout, we keep a folded code distinct from a broken one: the folded networks build a reproducible map of the fundamental domain and merely forget which copy of it they are in.

Results

Figure 1**Overview.** (**a**) The pipeline: an agent moves through a symmetric arena under actions drawn from a five-dimensional SpeedHD vector (Methods); a predictive RNN (pRNN) receives the current $7 \times 7 \times 3$ egocentric observation o_t and predicts the observation k steps ahead, $\hat{o}_{t+k}$. (**b**) The three arenas used throughout, named by their landmark symmetry group: C_1 (four distinct landmarks, no symmetry), C_2 (one landmark pair repeated under $180°$ rotation), and C_4 (one landmark repeated four times under $90°$ rotation); dashed outline marks a fundamental domain. (**c**) The animal phenomenon this study explains: Spiers et al. (2) recorded a high degree of spatial repetition across four visually identical compartments in a row; Grieves et al. (3) showed the same compartments arranged in parallel produce repeated place fields, while arranging them radially, so that a rotation rather than a translation relates them, produces remapping. (**d**) The central object of the Theorem: a group G (here C_4) partitions the state space X into orbits; the quotient map π sends each orbit to a single point of X/G, discarding which physically distinct member of the orbit the agent occupies.figure.caption.1 previews the elements this section formalises: the predictive-learning pipeline, the three landmark arenas, the animal phenomenon under explanation, and the quotient map at the centre of the argument.

The phenomenon: identical rooms fold under translation and separate under rotation. The cleanest place to see the idea is the geometry in which repeated place fields were first studied in rodents. We built two closed 6×6 compartments joined by a corridor and gave them identical interior landmarks, so the two rooms produced the same set of egocentric views: across all 144 combinations of location and heading the observations coincided exactly. In one arrangement the rooms are related by a translation, under which the compass reads the same in both; in the other by a rotation, un-



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der which it does not. We trained networks on each arrangement, decoded which room the agent occupied, and measured whether individual units repeated their firing fields between the rooms (Fig. 8**Functional consequences of the fold.** (**a**) Orbit-phase decoding accuracy vs prediction horizon k in the C_2 arena (chance 0.5); the non-invariant encodings (*full*, *parity*) resolve the fold at $k \geq 1$, the invariant ones (*axis*, *const*) stay near chance. (**b**) Coverage of the self-generated offline trajectory as a fraction of a wake path of equal length (dashed line, wake). Both step up at $k = 1$ and are ordered by how much of the compass the encoding keeps; points are mean $\pm$ s.e.m., $n = 6$. (**c**) Place-field repetition across two observationally identical compartments ($n = 8$ each): under translation the fields repeat and the room barely decodes (0.515 against a chance of 0.5; the residual is small but reliable), under rotation the fields do not repeat and the room decodes perfectly; within-room position (R^2) stays high in both. Dots are individual networks.figure.caption.13c).

The two arrangements gave opposite results ($n = 8$ networks each). Under translation the fields repeated almost exactly across the rooms (repetition 0.997) and the room was very nearly undecodable even by a nearest-neighbour classifier tested on cells it had already seen: 0.515 against a chance of 0.5. That residual is small, but it is real rather than noise ($p = 0.008$, Wilcoxon, $n = 8$), and we neither round it away nor lean on it. It is the incomplete fold we return to in the

Discussion, where its size and cause are the subject. On any reading the two rooms are very close to a single map. Under rotation the repetition vanished (repetition -0.080) and the room decoded perfectly (0.998). Position within a room decoded well in both arrangements ($R^2 = 0.95$ and 0.79), so neither network had stopped representing space. Every translation network separated from every rotation network on all four measures (Mann–Whitney $p = 1.6 \times 10^{-4}$, the exact two-sided floor at $n = 8$ vs. 8). Grieves et al. (3) reported this same contrast in CA1, and here it follows from a single fact about how the compass transforms: translation-invariant, rotation-equivariant (Methods; Limitations). The dissociation is visible directly in the population geometry, with no decoder: an Isomap embedding of the wake population state lands room B almost on top of room A under translation and pulls them an order of magnitude apart under rotation (Supplementary Information, Fig. S12).

The fold scales to the full four-compartment maze that Spiers et al. (2) recorded. We built four identical closed rooms in a row, each a translation of the last, off a shared corridor, with every view identical across the four rooms at matched local position and heading (verified exactly: zero pixel mismatch over every room-interior cell and heading, Methods). A predictive network ($n = 8$) folded all four onto very nearly a single map (Fig. 2**The predictive model reproduces place-field repetition.** (**a**) The four-room maze of Spiers et al. (2), model ($n = 8$): four translation-related identical rooms fold onto one map. The room decodes at 0.262 by a generalising decoder and 0.268 by a nearest-neighbour classifier at cells it has already seen, against a chance of 0.25 (dashed): a small residual, but a reliable one ($p = 0.008$). Fields repeat across the rooms (repetition 0.99) and within-room position stays intact ($R^2 = 0.96$). (**b**) A predictive network trained on a city-block maze shows the directional-repetition signature, predicted before any comparison to animal data: the directional index of a field (i) against that of its same-orientation partner (j) is positively correlated ($r = 0.62$; scatter subsampled for display, statistic from all pairs). Line, least-squares fit. An exploratory, non-diagnostic reanalysis of real CA1 data testing the same prediction is in the Supplementary Information (Fig. S10).figure.caption.2a): against the 0.25 chance of four alternatives, a generalising decoder recovered the room at 0.262 and a nearest-neighbour classifier tested on cells it had already visited at 0.268. As in the two-room maze, those residuals are small but not nothing ($p = 0.008$ for both, Wilcoxon, $n = 8$), and we report them as such. Fields repeated across the four rooms almost perfectly (repetition 0.994) and within-room position stayed intact ($R^2 = 0.96$). Room identity is decoded only from room-interior states; the connecting corridor is excluded for the reason given in Limitations. The two-room dissociation is thus not a small-n artefact: at the rodent room count, translation-related identical rooms collapse completely, the "high degree of spatial repetition" that Spiers et al. (2) observed.

The account makes a further prediction, which we test in the model before asking whether it is visible in real neurons. A city-block maze whose alleys run in two orthogonal orientations relates same-orientation alleys by translation (the compass reads the same) and orthogonal alleys by a $90°$ rotation, so a repeating cell's fields should share a directional preference across same-orientation alleys and not across orthogonal ones. A predictive network trained on this maze confirms it: the directional index of same-orientation field pairs is positively correlated ($r = 0.62$; Fig. 2**The predictive model reproduces place-field repetition.** (**a**) The four-room maze of Spiers et al. (2), model ($n = 8$): four translation-related identical rooms fold onto one map. The room decodes at 0.262 by a generalising decoder and 0.268 by a nearest-neighbour classifier at cells it has already seen, against a chance of 0.25 (dashed): a small residual, but a reliable one ($p = 0.008$). Fields repeat across the rooms (repetition 0.99) and within-room position stays intact ($R^2 = 0.96$). (**b**) A predictive network trained on a city-block maze shows the directional-repetition signature, predicted before any comparison to animal data: the directional index of a field (i) against that of its same-orientation partner (j) is positively correlated ($r = 0.62$; scatter subsampled for display, statistic from all pairs). Line, least-squares fit. An exploratory, non-diagnostic reanalysis of real CA1 data testing the same prediction is in the Supplementary Information (Fig. S10).figure.caption.2b), and this is the model's own prediction, not a fit to animal data. We then asked whether the signature is visible in real neurons, reanalysing the CA1 recordings of Hockeimer et al. (11). It is present, at a smaller magnitude ($r = 0.23$; 127 pairs, 40 cells, 5 rats), but the dataset cannot separate it from a conjunctive place-by-direction code that would produce the same correlation without any folding; two controls make the ambiguity explicit (Supplementary Information, Fig. S10). We therefore report the CA1 correlation as *consistent with* the quotient but not diagnostic of it, and treat the model prediction, not the animal reanalysis, as the load-bearing result of this section.

Why a predictive map must fold: the quotient theorem. The phenomenon has an explanation that needs no trained network at all. Return to the two identical rooms. A trajectory through one room and the corresponding trajectory through its symmetry-related copy generate, step for step, the same sequence of observations and the same sequence of encoded actions. That is what "identical rooms" and "an invariant compass" mean together. A network is a fixed function of that input sequence. If two histories feed it identical inputs, it must land in identical internal states, and no reading of those states can afterwards recover which room the agent was really in. The information was never in the stream. Everything below is this observation made precise.

Theorem (folding onto the quotient)**.** *Let a finite group G act freely on the arena's positions X, with a G-invariant start distribution, a G-equivariant observation likelihood $p(o \mid x)$, and a policy measurable with respect to the agent's belief. If the head-direction encoding is G-invariant, $\phi(h+g) = \phi(h)$ for every $g \in G$, then the posterior over position given any history of observations and encoded actions is G-invariant, and any decoder of the orbit element has accuracy at most $1/|G|$.*

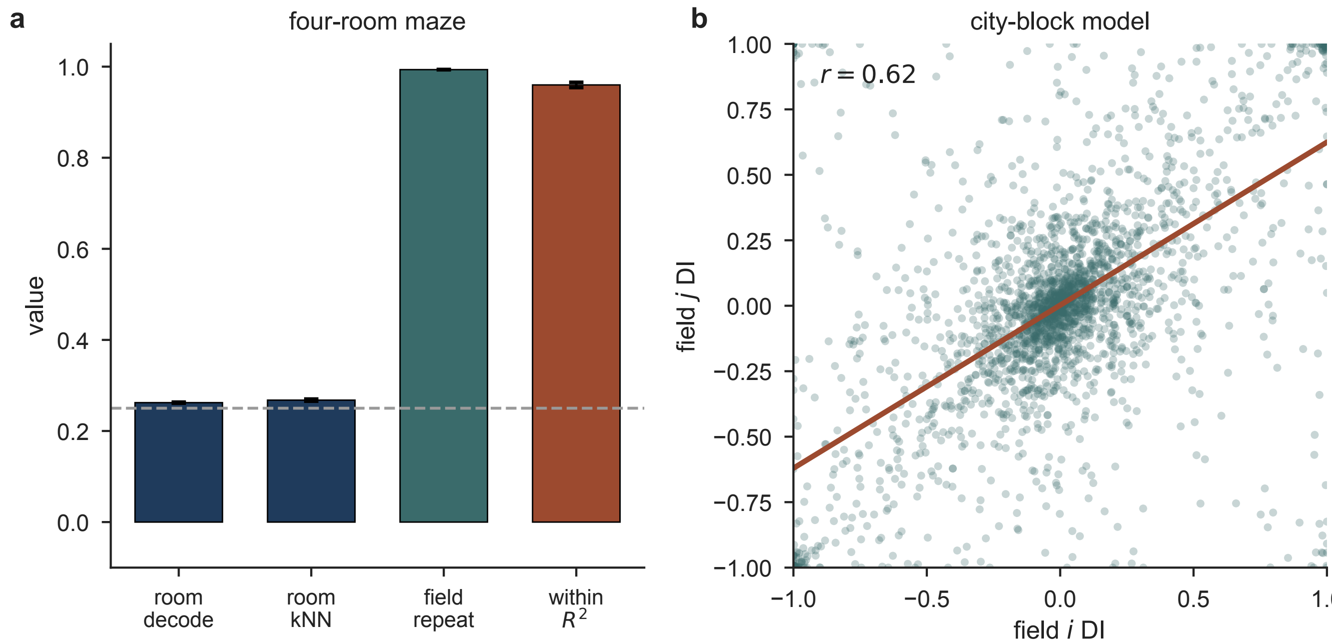


Fig. 2. The predictive model reproduces place-field repetition. (**a**) The four-room maze of Spiers et al. (2), model ($n = 8$): four translation-related identical rooms fold onto one map. The room decodes at 0.262 by a generalising decoder and 0.268 by a nearest-neighbour classifier at cells it has already seen, against a chance of 0.25 (dashed): a small residual, but a reliable one ($p = 0.008$). Fields repeat across the rooms (repetition 0.99) and within-room position stays intact ($R^2 = 0.96$). (**b**) A predictive network trained on a city-block maze shows the directional-repetition signature, predicted before any comparison to animal data: the directional index of a field (i) against that of its same-orientation partner (j) is positively correlated ($r = 0.62$; scatter subsampled for display, statistic from all pairs). Line, least-squares fit. An exploratory, non-diagnostic reanalysis of real CA1 data testing the same prediction is in the Supplementary Information (Fig. S10).

Proof. The action of $g \in G$ on trajectories, $\gamma \mapsto g \cdot \gamma$, is a measure-preserving bijection: it fixes the invariant start distribution and the equivariant transition and policy, and it leaves the sequence of observations and encoded actions unchanged, because G-related positions are observationally identical and ϕ is G-invariant. Histories ending at x and at $g \cdot x$ therefore induce the same input sequence, so the network map, a fixed function of that sequence, assigns them the same state, and the position posterior is constant on each G-orbit. No function of the state separates orbit-mates, so an $|G|$-way phase decoder is bounded by $1/|G|$.

The policy hypothesis is concrete, not abstract, and worth cashing out because it is easy to violate by accident. Every trajectory in this paper is generated by a random walk with fixed action probabilities (forward 0.60, turn left 0.15, turn right 0.15, stop 0.10; Methods), the same at every position, heading, and copy of the fundamental domain. Such a policy is a function of nothing the agent's belief could fail to represent, so it is trivially measurable with respect to that belief. The hypothesis is not vacuous: it would fail for a policy that consulted information outside the belief state, such as a script that explored one room more thoroughly than its g·room image, which would leak room identity into the action stream itself. Nothing in the experiments does this; the walk cannot tell the rooms apart any better than the compass can.

The bound is an upper limit, not a prediction. It forbids exceeding the ceiling but does not force the network to reach it, and the free-action assumption (no rotation fixes a cell, satisfied by the even grid) is what gives every orbit exactly $|G|$ members. Two things the paper turns on are therefore empirical, not entailed: that a *predictive* objective drives the code to *saturate* the bound, where a reconstruction objective leaves it far below (a later section), and that the folded code is folded rather than broken, since within-domain position survives. The quotient is pinched between a proved ceiling and an empirical floor.

The proof turns on nothing but the sigma-algebra the (observation, action) stream generates, so the ceiling is invariant to everything downstream of that stream, and this answers three worries at once. It does not reference the observation's format: a 7×7 pixel patch, a higher-resolution render, or a richer sensory channel are all G-equivariant whenever the arena is, so G-related histories stay indistinguishable whatever the observation's dimensionality. It does not reference network capacity: a larger or smaller decoder computes a function of the same aliased sequence, and no capacity manufactures information the sequence does not carry, so a bigger network reaches the ceiling faster, not past it. And it survives noise that leaves the equivariance intact. A compass corrupted on 30% of steps (a later section) changes how good each heading estimate is, not whether the encoding stays invariant, and the fold does not move. What lifts the ceiling is only a channel that breaks the equivariance itself: a statement about information content, not about richness, capacity, architecture, or noise.

The theorem also tells us how to *measure* the fold. Under folding onto X/G the population state satisfies $h(x) = h(R^2 x)$, so no decoder can recover which element of the orbit $\{x, R^2 x\}$ produced it. We therefore train a linear decoder for *orbit phase*, holding out whole orbits across cross-validation folds so the decoder must generalise a phase direction to positions it has never seen. Chance is 0.500; a fully folded code sits at the ceiling, an unfolded one well above it. This is the readout the rest of the paper leans on, and the reason it can be trusted where single-cell measures cannot is the subject of a later section.

A design that separates equivariance from information. The theorem hangs on one word, *invariance*, and the experiment has to hold everything else fixed to isolate it. Two hypotheses are on the table. If folding tracks how much the compass tells the network, then any impoverished compass should fold. If it tracks equivariance, then only a compass invariant under the arena's symmetry should fold, and one carrying the same information but not invariant should not. These make opposite predictions, and the way to force them apart is a pair of head-direction encodings matched in every respect but invariance.

That is what the four encodings do. Each network receives a $7 \times 7 \times 3$ egocentric view and a five-dimensional SpeedHD action vector, $[\text{speed}, \ \text{onehot}(h)]$ with $h \in \{\text{E,S,W,N}\}$; we replace the heading block by $M\,\text{onehot}(h)$ for a fixed 4×4 matrix M, preserving dimensionality and column sums (Table 1The four head-direction encodings. *axis* and *parity* are matched at one bit and differ only in invariance.table.caption.3, Fig. 3**Design and phenotype.** (**a**) The four head-direction encodings, as 4×4 transforms of the one-hot heading block $\{\text{E,S,W,N}\}$; grey level is the matrix weight. *full* is the identity (two bits); *parity* and *axis* collapse the compass to one bit with different partitions; *const* discards it. *axis* and *const* are invariant under the 180° rotation and fold the C_2 arena; *const* is the only encoding invariant under the full C_4 group. *axis* and *parity* carry identical entropy (one bit) and differ only in invariance. (**b**) Mean firing-rate maps (smoothed, peak-normalised) of example units. Fields are single and sharp when the code does not fold, and repeat at the symmetry-related locations, once under C_2 (*axis* encoding) and four times under C_4 (*const*). Units are representative; the complete unselected set for each condition is given in the Supplementary Information.figure.caption.4a). *full* keeps all four headings (two bits). *const* discards them (zero bits). The crucial pair is in between: *axis* and *parity* each collapse the compass to a single bit, with identical entropy and identical activation statistics. They differ only in *which* headings they group together, and so in whether the bit is invariant under the 180° rotation R^2. *axis* groups $\{E, W\}$ against $\{N, S\}$ and is invariant under R^2; *parity* groups $\{E, S\}$ against $\{W, N\}$ and is not. If information drives the fold, they behave alike. If invariance drives it, *axis* folds and *parity* does not. Nothing else about the two differs.

Predictive learning grows place cells, and symmetry makes them repeat. Trained on prediction alone, the networks develop units with localised spatial firing fields, the place cells reported for this objective by Levenstein et al. (1) (Fig. 3**Design and phenotype.** (**a**) The four head-direction encodings, as 4×4 transforms of the one-hot heading block $\{\text{E,S,W,N}\}$; grey level is the matrix weight. *full* is the identity (two bits); *parity* and *axis* collapse the compass to one bit with different partitions; *const* discards it. *axis* and *const* are invariant under the 180° rotation and fold the C_2 arena; *const* is the only encoding invariant under the full C_4 group. *axis* and *parity* carry identical entropy (one bit) and differ only in invariance. (**b**) Mean firing-rate maps (smoothed, peak-normalised) of example units. Fields are single and sharp when the code does not fold, and repeat at the symmetry-related locations, once under C_2 (*axis* encoding) and four times under C_4 (*const*). Units are representative; the complete unselected set for each condition is given in the Supplementary Information.figure.caption.4b). In the asymmetric arena under the full compass, about 480 of the 500 units carry a place field (a median of about two fields per unit). The effect of symmetry is already visible in the rate maps, before any decoder is applied. When the arena is symmetric and the head-direction encoding is invariant to that symmetry, the fields repeat: a unit in the C_2 arena under *axis* fires at a location and again at its 180° image, and a unit in the C_4 arena under *const* fires at all four 90° copies. A single unit reports every member of a symmetry orbit as the same place. The sections that follow quantify this and isolate what controls it.

Matched bits, opposite maps: equivariance is the cause. With the readout in hand and the encodings matched, the whole experiment fits in one table. Orbit-phase decoding separates the encodings exactly as invariance, not information, predicts (Table 2Orbit-phase decoding accuracy (chance $= 0.500$). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.table.caption.5, Fig. 4**Invariance, not information, folds the code.** (**a**) Orbit-phase decoding accuracy for every trained network (one dot per network; horizontal bar, group mean), by head-direction encoding and arena; chance 0.5 (dashed). *full* and *parity* stay high in every arena, whereas the invariant encodings *axis* and *const* fall to chance in the symmetric arenas. (**b**) The matched-entropy contrast: *axis* and *parity* carry one bit each and are indistinguishable in the C_1 arena (both ≈ 0.97) but separate completely in C_2 (*parity* 0.955, *axis* 0.552; every *parity* network scores above every *axis* network, Mann–Whitney $p = 1.1 \times 10^{-5}$). Data of Table 2Orbit-phase decoding accuracy (chance $= 0.500$). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.table.caption.5.figure.caption.6). *full* and *parity* stay high in every arena. The invariant encodings, *axis* and *const*, fall to chance in precisely the arenas whose symmetry they cannot break.

The decisive comparison is the matched pair. In the C_2 arena, *axis* and *parity* carry the same entropy (one bit) and the same information about the current heading, yet they differ by 0.403 in phase accuracy (0.552 against 0.955). What separates them is equivariance: *parity*'s bit is informative about orbit phase, since it distinguishes a position from its 180° image, whereas *axis*'s bit is invariant under that rotation and so carries none. The separation is complete: every *parity* network scores above every *axis* network (Mann–Whitney $U = 0$, $p = 1.1 \times 10^{-5}$, the exact two-sided floor at $n = 10$ vs. 10; $p = 3.2 \times 10^{-5}$ after Bonferroni correction across the three arenas). And in the C_1 arena, where there is no symmetry to collapse onto, the same two encodings are indistin-

Table 1. The four head-direction encodings. *axis* and *parity* are matched at one bit and differ only in invariance.

encoding	bits	C_2-invariant	C_4-invariant
full (E,S,W,N)	2.0	no	no
parity ({E,S} vs {W,N})	1.0	no	no
axis ({E,W} vs {N,S})	1.0	**yes**	no
const	0.0	yes	**yes**

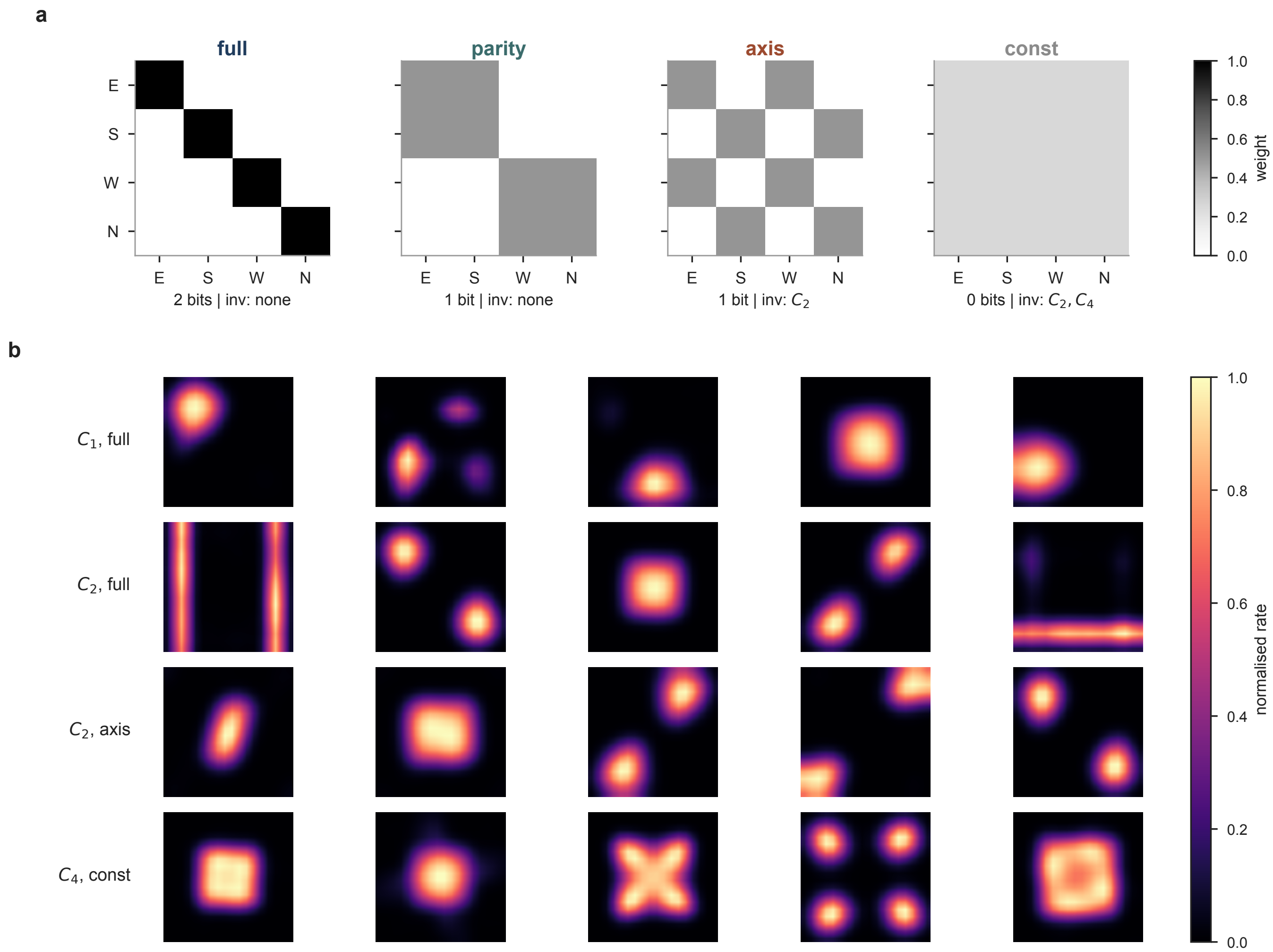


Fig. 3. Design and phenotype. (**a**) The four head-direction encodings, as 4×4 transforms of the one-hot heading block $\{$E,S,W,N$\}$; grey level is the matrix weight. *full* is the identity (two bits); *parity* and *axis* collapse the compass to one bit with different partitions; *const* discards it. *axis* and *const* are invariant under the 180° rotation and fold the C_2 arena; *const* is the only encoding invariant under the full C_4 group. *axis* and *parity* carry identical entropy (one bit) and differ only in invariance. (**b**) Mean firing-rate maps (smoothed, peak-normalised) of example units. Fields are single and sharp when the code does not fold, and repeat at the symmetry-related locations, once under C_2 (*axis* encoding) and four times under C_4 (*const*). Units are representative; the complete unselected set for each condition is given in the Supplementary Information.

Table 2. Orbit-phase decoding accuracy (chance $= 0.500$). $n = 16$ per cell for s1 and s2, $n = 8$ for s4.

arena	*full*	*parity*	*axis*	*const*
s1 (C_1, no symmetry)	0.981	0.972	0.971	0.966
s2 (C_2)	0.978	0.955	**0.552**	**0.552**
s4 (C_4)	0.984	0.933	**0.544**	**0.521**

guishable ($\Delta = 0.000$, $p = 0.97$). That C_1 null is what licenses the causal reading: *axis* does not degrade spatial coding in general; it degrades it exactly when the arena has a symmetry under which *axis* is invariant. The chance-level reading is not an artefact of the linear decoder. A k-nearest-neighbour decoder and a multi-layer perceptron also sit at chance for the invariant encodings, and recover phase for the non-invariant ones (Supplementary Information). The orbit-phase information is absent from the folded code, not merely hidden from a linear readout.

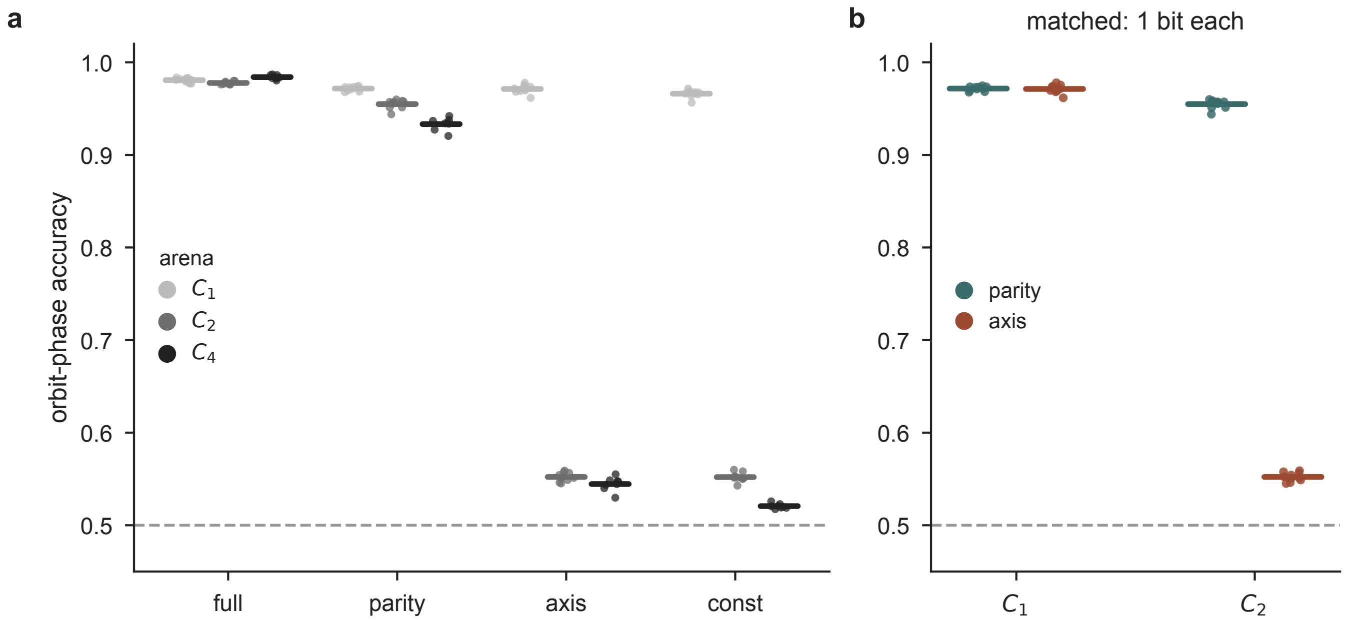


Fig. 4. Invariance, not information, folds the code. (**a**) Orbit-phase decoding accuracy for every trained network (one dot per network; horizontal bar, group mean), by head-direction encoding and arena; chance 0.5 (dashed). *full* and *parity* stay high in every arena, whereas the invariant encodings *axis* and *const* fall to chance in the symmetric arenas. (**b**) The matched-entropy contrast: *axis* and *parity* carry one bit each and are indistinguishable in the C_1 arena (both ≈ 0.97) but separate completely in C_2 (*parity* 0.955, *axis* 0.552; every *parity* network scores above every *axis* network, Mann–Whitney $p = 1.1 \times 10^{-5}$). Data of Table 2Orbit-phase decoding accuracy (chance $= 0.500$). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.table.caption.5.

The C_4 map hits the group-theoretic $1/|G|$ ceiling. The theorem makes a sharper prediction than "the code folds." A code folded onto X/G can name the coset but not the element within it, so a $|G|$-way phase decoder is bounded by exactly $1/|G|$, a number with no free parameters. The C_4 arena, read through its own four-way quotient (chance 0.250), is where this is testable, and the networks land on it (Table 3C_4 four-way phase decoding in the s4 arena (chance $= 0.250$); $n = 4$ networks per encoding.table.caption.7, Fig. 5**The C_4 arena reproduces the group-theoretic ceiling.** Four-way orbit-phase accuracy in the C_4 arena (one dot per network; bar, mean; chance 0.25). A code folded onto X/G can name the coset but not the orbit element, so a $|G|$-way decoder is bounded by $1/|G|$. *const*, invariant under the full C_4 group, falls to the $1/4$ ceiling (dotted); the C_2-invariant *axis* falls to the $1/2$ ceiling; *full* and *parity*, not invariant, stay high. Ceilings are the group-theoretic predictions, not fits. $n = 4$ networks per encoding.figure.caption.8). *const*, the only C_4-invariant encoding, falls to the C_4 floor (0.275, just above 0.250). *axis*, invariant under C_2 but not C_4, falls only to the C_2 ceiling of 0.5 (0.482). The non-invariant encodings stay high. The folded values sit a few percent above their predicted floors rather than exactly on them, and that residual is real signal rather than decoder bias: across networks, C_2 phase accuracy under *axis* is 0.552 against its floor of 0.500 (one-sample t-test, $t(9) = 34.8$, $p = 6.6 \times 10^{-11}$, $n = 10$), and under *const* 0.552 ($t(9) = 35.2$, $p = 6.0 \times 10^{-11}$). The fold is very close to, but not exactly, complete. The decisive quantity for the account is which floor each encoding falls to, not the residual's exact size (Discussion).

This also resolves what *const* in the C_4 arena looked like under the wrong readout. Read through the C_2 quotient it appeared dead; read through the C_4 quotient it is plainly a live spatial code, its C_4 fundamental-domain R^2 is $+0.646$ (minimum $+0.535$ across the four *const* networks), and its four-way phase accuracy of 0.275 sits at the C_4 chance floor. What looked like a dead code was a code folded onto a larger quotient than the readout was built to see. We flag this because a G-folded code and a code that has stopped representing space are easy to confuse, and only the second is a failure.

A folded map, not a broken one. Everything so far is a negative: a decoder that cannot recover orbit phase. A negative is a weak thing to build on, because a merely degraded code also fails a decoder. So we asked the positive question. If the quotient law is right, a folded code is not a poor map of the arena; it is a *faithful map of a different space*. That is a claim about geometry, and it can be tested directly.

The folded networks have not stopped representing space. For 104 of the 112 networks, $\max(R^2_{\mathrm{raw}}, R^2_{\mathrm{domain}}) \geq 0.81$: the folded codes still linearly encode position *within the fundamental domain*, and have merely discarded which copy of it the agent occupies (Fig. 6**The fold, seen geometrically.** (**a**) The fold ratio: the distance between a cell's population state and that of its 180° image, in units of the distance to an adjacent cell. One dot per network, $n = 10$ per condition; bar, mean; whiskers, 95% bootstrap interval. Below one, orbit partners sit closer together than spatial neighbours do, which is what folding means. Only *axis* in the C_2 arena is below one (0.457 ± 0.004); the same encoding with no symmetry to fold onto is not (*axis*/C_1, 1.872), nor is *parity*/C_2 (2.083), nor *full*/C_2 (2.675). The ratio is computed in the full hidden space, so it is not an artefact of the projection used to plot it: the folded case reads below one and every non-folded

Table 3. C_4 four-way phase decoding in the s4 arena (chance $= 0.250$); $n = 4$ networks per encoding.

encoding	invariance	measured	predicted ceiling
full	none	0.918	—
parity	none	0.860	—
axis	C_2 only	**0.482**	0.500
const	C_2 and C_4	**0.275**	0.250

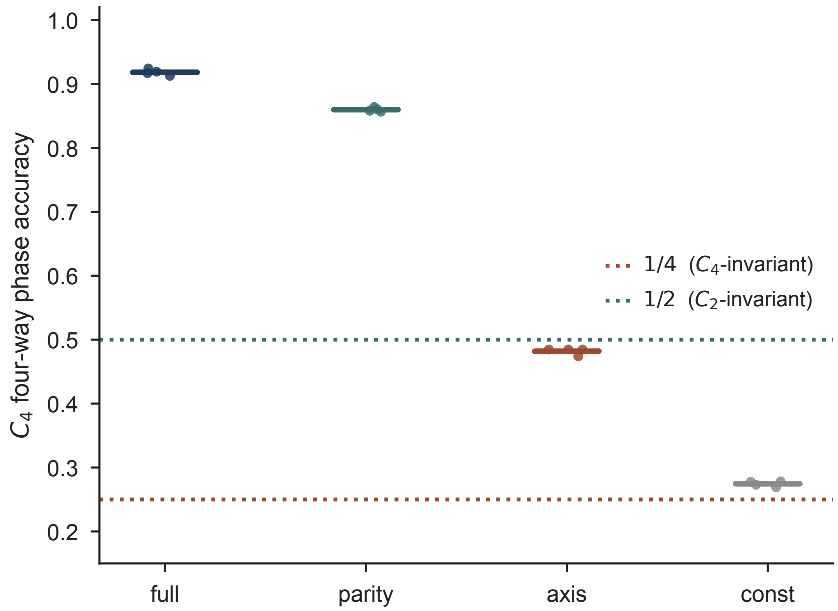


Fig. 5. The C_4 arena reproduces the group-theoretic ceiling. Four-way orbit-phase accuracy in the C_4 arena (one dot per network; bar, mean; chance 0.25). A code folded onto X/G can name the coset but not the orbit element, so a $|G|$-way decoder is bounded by $1/|G|$. *const*, invariant under the full C_4 group, falls to the $1/4$ ceiling (dotted); the C_2-invariant *axis* falls to the $1/2$ ceiling; *full* and *parity*, not invariant, stay high. Ceilings are the group-theoretic predictions, not fits. $n = 4$ networks per encoding.

case above one under nonlinear embeddings (Isomap, t-SNE) as well as PCA. (**b**, **c**) Position-conditioned population manifold of the most- and the least-folded condition (one point per arena cell, PCA to three dimensions, coloured by C_2 orbit phase), illustrative; the fold ratio in (a) is the quantitative claim. The two phases superimpose under *axis*/C_2 and occupy separate regions under *full*/C_2. We reduce with PCA because the claim is metric, that two positions coincide, and unlike neighbour-graph embeddings PCA preserves distances rather than reshaping them.figure.caption.9). The eight exceptions are *const* in the C_4 arena, which fails both C_2-based readouts ($R^2_{\text{raw}} = -0.069$, $R^2_{\text{domain}} = -0.589$) not because it has stopped coding space but because it is folded by the full C_4 group, which a C_2 decoder cannot see. Domain position and orbit phase move independently across every network and encoding (Supplementary Information, Fig. S11), which is the coset-versus-phase signature of a fold rather than a dead code.

To test the positive claim we took, for each network, the population state averaged at every arena cell and measured the geodesic distances between those states. We then asked which of two candidate metrics those neural distances reproduce: the arena's own metric d_X, or the quotient metric $d_{X/G}(x,y) = \min_{g\in G}\|x - g\cdot y\|$. Fit is Kruskal stress at an optimal scale, since neural units are arbitrary and an isometry is only ever defined up to a scale factor (Methods).

The head-direction encoding decides which space the manifold is a map of, and it does so with one systematic exception we report rather than round away. Taking for each network whichever of d_X, d_{X/C_2} and d_{X/C_4} fits its geometry best, 103 of 112 networks are assigned by their encoding alone: *full* and *parity* map X; *axis* maps X/C_2; *const* maps X/C_4. The nine that are not are almost all one cell: every *axis* network in the C_4 arena (8/8) is fitted better by X/C_4 than by the X/C_2 its encoding predicts. This disagrees with our decoder, which puts the same networks at 0.482, the C_2 ceiling. The decoder is the measure we trust here, because $d_{X/G} \leq d_X$ pointwise, so a *larger* group can always fit a compressed code better, biasing the stress comparison toward over-folding in exactly this nested case. We flag the disagreement, we do not resolve it here, and no claim in this paper rests on the geometry metric where it and the decoder part company.

Where they agree, the geometry is unambiguous. In the C_2 arena the folded *axis* code fits the quotient at stress 0.190 against 0.468 for the arena, the mirror image of *full*, which fits the arena at 0.097 against 0.485 for the quotient (Fig. 7**The folded manifold is a map of the quotient.** One dot per network ($n = 10$ per condition, 8 for C_4); stress is Kruskal stress-1 at optimal scale, lower is a better fit. (**a**) Fit to the arena metric d_X against fit to the quotient metric d_{X/C_2}, with the line of equal fit. *full* and *parity* lie above it,

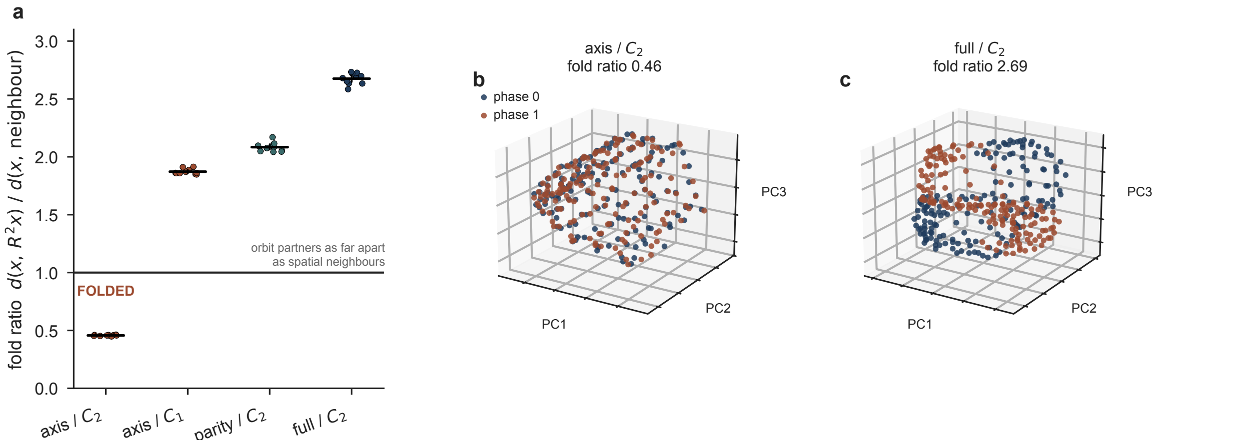


Fig. 6. The fold, seen geometrically. (**a**) The fold ratio: the distance between a cell's population state and that of its 180° image, in units of the distance to an adjacent cell. One dot per network, $n = 10$ per condition; bar, mean; whiskers, 95% bootstrap interval. Below one, orbit partners sit closer together than spatial neighbours do, which is what folding means. Only *axis* in the C_2 arena is below one (0.457 ± 0.004); the same encoding with no symmetry to fold onto is not (*axis*/C_1, 1.872), nor is *parity*/C_2 (2.083), nor *full*/C_2 (2.675). The ratio is computed in the full hidden space, so it is not an artefact of the projection used to plot it: the folded case reads below one and every non-folded case above one under nonlinear embeddings (Isomap, t-SNE) as well as PCA. (**b**, **c**) Position-conditioned population manifold of the most- and the least-folded condition (one point per arena cell, PCA to three dimensions, coloured by C_2 orbit phase), illustrative; the fold ratio in (a) is the quantitative claim. The two phases superimpose under *axis*/C_2 and occupy separate regions under *full*/C_2. We reduce with PCA because the claim is metric, that two positions coincide, and unlike neighbour-graph embeddings PCA preserves distances rather than reshaping them.

so they are better described as maps of X; *axis* lies below it, so it is better described as a map of X/C_2. Dotted line, the stress returned when the code's spatial content is destroyed by shuffling, a reachable ceiling. (**b**) The control for the compression artefact: $d_{X/G} \leq d_X$ pointwise, so any group shrinks long distances. A sham order-2 group (translation by half the arena) compresses identically but is not a symmetry, and the folded code prefers the true C_2 over it by 0.343. (**c**) The same claim without any metric: the cosine between the population state at x and at its 180° image. Under *axis* in the C_2 arena it is 0.990, so the orbit is one point; under *parity*, which carries the same one bit, it is 0.741.figure.caption.10a). The compass does not degrade the map; it selects the space the map is of. One artefact had to be excluded first. Because $d_{X/G} \leq d_X$ pointwise, any group shrinks long distances, so a code whose neural distances merely saturate at long range would prefer a quotient metric for reasons unrelated to symmetry. We therefore built a sham group of the same order: a translation by half the arena, which compresses distances exactly as C_2 does but is a symmetry of no arena here. The folded code prefers the true symmetry over the sham by 0.343 (Fig. 7**The folded manifold is a map of the quotient.** One dot per network ($n = 10$ per condition, 8 for C_4); stress is Kruskal stress-1 at optimal scale, lower is a better fit. (**a**) Fit to the arena metric d_X against fit to the quotient metric d_{X/C_2}, with the line of equal fit. *full* and *parity* lie above it, so they are better described as maps of X; *axis* lies below it, so it is better described as a map of X/C_2. Dotted line, the stress returned when the code's spatial content is destroyed by shuffling, a reachable ceiling. (**b**) The control for the compression artefact: $d_{X/G} \leq d_X$ pointwise, so any group shrinks long distances. A sham order-2 group (translation by half the arena) compresses identically but is not a symmetry, and the folded code prefers the true C_2 over it by 0.343. (**c**) The same claim without any metric: the cosine between the population state at x and at its 180° image. Under *axis* in the C_2 arena it is 0.990, so the orbit is one point; under *parity*, which carries the same one bit, it is 0.741.figure.caption.10b), while the unfolded codes prefer neither. The preference is for the group, not for the compression. The same claim survives with no metric at all: the cosine between the population state at x and at its 180° image is 0.990 under *axis* and 0.741 under *parity* in the C_2 arena (Mann–Whitney $U = 100$, $p = 1.1 \times 10^{-5}$; Fig. 7**The folded manifold is a map of the quotient.** One dot per network ($n = 10$ per condition, 8 for C_4); stress is Kruskal stress-1 at optimal scale, lower is a better fit. (**a**) Fit to the arena metric d_X against fit to the quotient metric d_{X/C_2}, with the line of equal fit. *full* and *parity* lie above it, so they are better described as maps of X; *axis* lies below it, so it is better described as a map of X/C_2. Dotted line, the stress returned when the code's spatial content is destroyed by shuffling, a reachable ceiling. (**b**) The control for the compression artefact: $d_{X/G} \leq d_X$ pointwise, so any group shrinks long distances. A sham order-2 group (translation by half the arena) compresses identically but is not a symmetry, and the folded code prefers the true C_2 over it by 0.343. (**c**) The same claim without any metric: the cosine between the population state at x and at its 180° image. Under *axis* in the C_2 arena it is 0.990, so the orbit is one point; under *parity*, which carries the same one bit, it is 0.741.figure.caption.10c). Under the invariant encoding the orbit has collapsed to a point; under the encoding carrying the same one bit it has not. The same collapse appears as a glued fundamental domain in a nonlinear Isomap embedding: a position and its 180° image sit at 0.17–0.18 of a random pair's distance under *axis*, against 0.85–0.89 under *full* (Supplementary Information, Fig. S8). It is a property of the geometry, not of the projection used to see it.

The geometry also separates the theorem's two conditions,

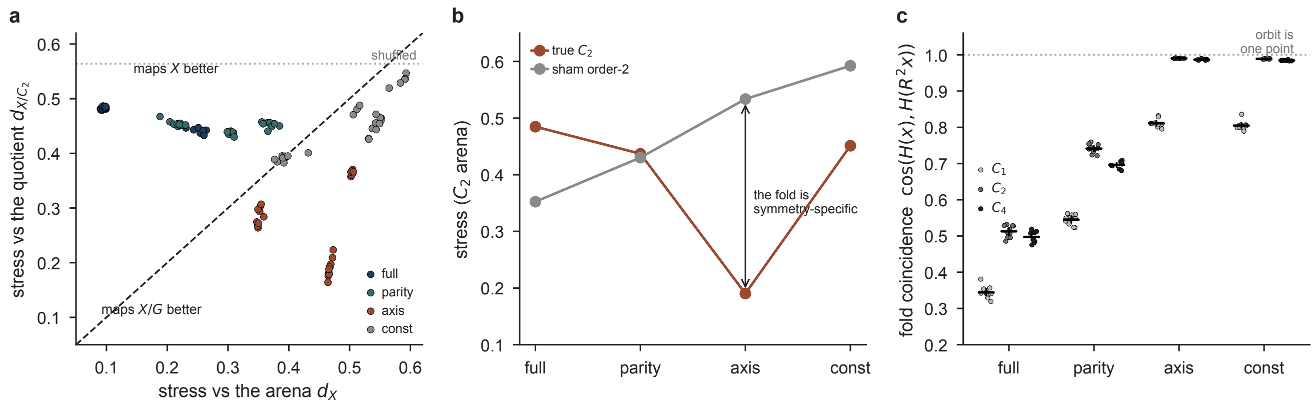


Fig. 7. The folded manifold is a map of the quotient. One dot per network ($n = 10$ per condition, 8 for C_4); stress is Kruskal stress-1 at optimal scale, lower is a better fit. (**a**) Fit to the arena metric d_X against fit to the quotient metric d_{X/C_2}, with the line of equal fit. *full* and *parity* lie above it, so they are better described as maps of X; *axis* lies below it, so it is better described as a map of X/C_2. Dotted line, the stress returned when the code's spatial content is destroyed by shuffling, a reachable ceiling. (**b**) The control for the compression artefact: $d_{X/G} \le d_X$ pointwise, so any group shrinks long distances. A sham order-2 group (translation by half the arena) compresses identically but is not a symmetry, and the folded code prefers the true C_2 over it by 0.343. (**c**) The same claim without any metric: the cosine between the population state at x and at its 180° image. Under *axis* in the C_2 arena it is 0.990, so the orbit is one point; under *parity*, which carries the same one bit, it is 0.741.

which the decoder cannot. *axis* folds the *geometry* by C_2 even in the C_1 arena, where there is no symmetry at all (it prefers d_{X/C_2} over the sham by 0.145, and its orbit cosine is 0.811), and yet orbit phase there stays decodable at 0.971. An invariant compass pulls symmetry-related states together wherever it is used; it is the *observations* being equivariant that also destroys the information. Only where both hold does the code fold completely, which is the intersection G the introduction promised, now seen in the geometry.

Prediction builds the fold; reconstruction does not. The fold, so far, is a property of the inputs. Whether a network actually builds it turns out to depend on what the network is trained to do, and this is the one manipulation that could have shown the fold is an accident of the architecture rather than a consequence of prediction.

Start with the inputs alone. Call two states (x, h) and (x', h') *input-equivalent* if they produce the same 7×7 observation and the same HD encoding. A code is a function of its inputs, so no decoder can separate input-equivalent states, and phase decoding is bounded above by

$$\mathrm{acc}_{\max} = \tfrac{1}{2} + \tfrac{1}{2} \Pr_{(x,h)} \big[(x,h) \not\sim (R^2 x, h+2) \big], \qquad \textbf{(1)}$$

the fraction of orbit pairs the input distinguishes. Evaluating the right-hand side on the arenas themselves, by enumerating all 324×4 states and hashing their observations, takes no training at all, and it predicts the trained networks to a mean absolute error of 0.031 across the twelve cells (Table 4The bound of Eq. 1equation.0.1, computed from the arenas with no free parameters, against the trained networks. Every encoding falls to the accuracy its input-distinguishability predicts; the mean absolute error across the twelve cells is 0.031. (We report the bound as a bound, not a fit: the predicted values are near-degenerate, clustered at 1.0 and at the folding floors, so a correlation coefficient across cells would merely register that separation.)table.caption.11). It also settles a number that would otherwise look like a failure: in the C_1 arena *axis* reaches 0.971 rather than 1.000 because 6.6% of states there are *accidentally* observationally identical to their C_2 partner, putting the bound at 0.967. *axis* and *const* sit exactly on it. They were not failing to fold; they were folding on precisely the ambiguous fraction.

The bound is an upper limit, and a network is free to discard information its loss does not reward. To see whether the objective is what recruits the compass, we held the wiring, the compass, and the training data fixed and changed only the rollout horizon k. At $k = 0$ the target is the network's own current observation: a pure autoencoder, with the same architecture and the same compass. At $k \ge 1$ it must predict a future observation. The result is a switch (Table 5Orbit-phase decoding in the C_2 arena as a function of prediction horizon k (chance $= 0.500$, $n = 6$ per cell).table.caption.12, Fig. 8**Functional consequences of the fold.** (**a**) Orbit-phase decoding accuracy vs prediction horizon k in the C_2 arena (chance 0.5); the non-invariant encodings (*full*, *parity*) resolve the fold at $k \ge 1$, the invariant ones (*axis*, *const*) stay near chance. (**b**) Coverage of the self-generated offline trajectory as a fraction of a wake path of equal length (dashed line, wake). Both step up at $k = 1$ and are ordered by how much of the compass the encoding keeps; points are mean $\pm$ s.e.m., $n = 6$. (**c**) Place-field repetition across two observationally identical compartments ($n = 8$ each): under translation the fields repeat and the room barely decodes (0.515 against a chance of 0.5; the residual is small but reliable), under rotation the fields do not repeat and the room decodes perfectly; within-room position (R^2) stays high in both. Dots are individual networks.figure.caption.13a). At $k = 0$ the input still separates 1228 of the 1296 states under *full*, so the compass is right there, yet all four encodings collapse to within 0.005 of one another, landing on the *observation-only* bound of 0.500; the spread across encodings is 0.0052 at $k = 0$ against 0.427 at $k = 5$. The compass is not absent under reconstruction; it is *discarded*: reconstructing o_t never requires knowing which of two identical places you occupy, whereas predicting o_{t+1}

Table 4. The bound of Eq. 1equation.0.1, computed from the arenas with no free parameters, against the trained networks. Every encoding falls to the accuracy its input-distinguishability predicts; the mean absolute error across the twelve cells is 0.031. (We report the bound as a bound, not a fit: the predicted values are near-degenerate, clustered at 1.0 and at the folding floors, so a correlation coefficient across cells would merely register that separation.)

arena	encoding	distinguishable	predicted	measured
s1	*full / parity*	1.000	1.000	0.981 / 0.972
s1	*axis / const*	0.934	0.967	0.971 / 0.966
s2	*full / parity*	1.000	1.000	0.978 / 0.955
s2	*axis / const*	0.000	0.500	0.552 / 0.552
s4	*full / parity*	1.000	1.000	0.984 / 0.933
s4	*axis / const*	0.000	0.500	0.544 / 0.521

does, and one step is enough. Everything downstream, the ceiling, the replay, and the lesion effects, is therefore not a property of a network that happens to also predict. It is what a fixed architecture does *because* the objective is prediction.
The same switch appears in place-field size, and it matters for a common reading of the predictive horizon. If k behaved like a successor-representation discount, place fields should widen smoothly with it, giving the kind of representation-scale gradient reported along the hippocampal dorsoventral axis (12). Instead mean field area (*full*, C_2) jumps once and stops: 33.8 ± 1.6 cells at $k = 0$, 40.7 ± 1.0 at $k = 1$, and no further (42.1 ± 1.0 at $k = 3$, 39.7 ± 0.7 at $k = 5$; `field_area_horizon.csv`). Prediction versus no prediction sets field size once, the same way it sets the fold; how far ahead the network predicts does not additionally scale it, at least across this range of k and at fixed hidden size. This architecture does not offer the discount-gradient account of the dorsoventral axis for free.

Map and replay are the same variable, read two ways. A predictive network can be run offline, driven by its own predicted observation, to generate trajectories without moving through the arena (1). Its offline behaviour tracks the fold. Both the coverage of the arena (relative to a waking path of equal length) and the sequenceness of the rollout (the rank correlation between principal-axis position and time, against time-shift and cell-shuffle nulls) step up between $k = 0$ and $k = 1$, the same range as orbit-phase accuracy, and both fall as the compass is weakened (Fig. 8**Functional consequences of the fold.** (**a**) Orbit-phase decoding accuracy vs prediction horizon k in the C_2 arena (chance 0.5); the non-invariant encodings (*full*, *parity*) resolve the fold at $k \geq 1$, the invariant ones (*axis*, *const*) stay near chance. (**b**) Coverage of the self-generated offline trajectory as a fraction of a wake path of equal length (dashed line, wake). Both step up at $k = 1$ and are ordered by how much of the compass the encoding keeps; points are mean $\pm$ s.e.m., $n = 6$. (**c**) Place-field repetition across two observationally identical compartments ($n = 8$ each): under translation the fields repeat and the room barely decodes (0.515 against a chance of 0.5; the residual is small but reliable), under rotation the fields do not repeat and the room decodes perfectly; within-room position (R^2) stays high in both. Dots are individual networks.figure.caption.13b). Coverage at $k = 1$ runs 1.30, 0.80, 0.45 and 0.31 times waking under *full*, *parity*, *axis* and *const*, the order in which the code folds, and sequenceness falls from 0.53 under *full* (waking 0.56; its own cell-shuffle floor 0.26) to 0.21 under *axis* (floor 0.17), approaching that floor without reaching it.
One caveat belongs beside the coverage numbers, not in a supplement. A shuffled rollout covers $1.57\times$ the waking path, more than *any* of our networks, the best of which reaches $1.30\times$. Coverage rewards incoherence: a trajectory that teleports visits more of the arena than one that walks. It is a measure of extent, not of structure, and cannot on its own evidence structured replay. Sequenceness is the readout that carries that claim, and it is the one with a null the real networks clear (*full* 0.53 against a floor of 0.26). We report coverage because the $k = 0 \to k = 1$ step is real and matters, and we report its null because it is unflattering.
This is more than a shared horizon; it is the same variable measured twice. The k-sweep that takes the map from the observation-only bound to a resolved map (Table 5Orbit-phase decoding in the C_2 arena as a function of prediction horizon k (chance $= 0.500$, $n = 6$ per cell).table.caption.12) is the identical manipulation that takes offline coverage from $0.37\times$ waking under the autoencoder to full engagement under one step of prediction, and the *full*/*parity*/*axis*/*const* order in which encodings fold the map is the order in which they lose the replay. A predictive objective does not separately grow a spatial map and a capacity to generate offline activity. It grows one thing, self-motion integration, recruited because prediction requires it, and the map and the replay are two readouts of it. Reconstruction, the only objective tested here that removes the mechanism, removes both readouts at once and by the same amount. This is the sense in which predictive learning is a unifying account of the spatial code rather than a fact about place-field statistics: not that it produces a map and, separately, produces replay, but that the one thing it does is read out as both.
Two limits on the replay claim. Sequenceness is scored as $|\mathrm{Spearman}(x(t), t)|$, an absolute value, so it measures whether a rollout is an ordered sweep at all and is blind by construction to whether that sweep runs forward or in reverse. We therefore make no claim here about forward versus reverse replay; real reverse replay is reward-triggered (13, 14) and this network has no reward signal, so the account predicts its sweeps should not show that reward-linked form, but testing it needs a directional statistic we have not built.

The fold survives a noisy compass and arises from a learned one. The head-direction signal so far is a noise-

Table 5. Orbit-phase decoding in the C_2 arena as a function of prediction horizon k (chance $= 0.500$, $n = 6$ per cell).

k	*full*	*parity*	*axis*	*const*	*axis* vs *parity*
0 (autoencoder)	0.536	0.541	0.539	0.539	$p = 0.59$
1	0.975	0.941	0.545	0.546	$p = 0.00216$
3	0.968	0.947	0.548	0.545	$p = 0.00216$
5	0.978	0.953	0.553	0.551	$p = 0.00216$

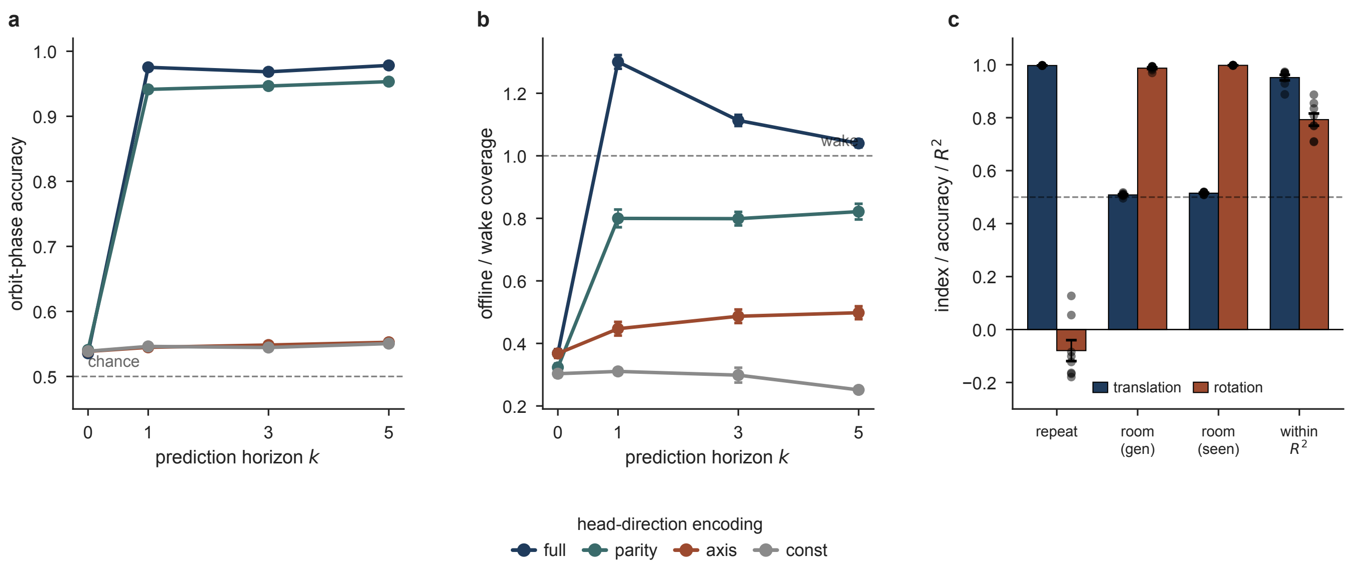


Fig. 8. Functional consequences of the fold. (**a**) Orbit-phase decoding accuracy vs prediction horizon k in the C_2 arena (chance 0.5); the non-invariant encodings (*full*, *parity*) resolve the fold at $k \geq 1$, the invariant ones (*axis*, *const*) stay near chance. (**b**) Coverage of the self-generated offline trajectory as a fraction of a wake path of equal length (dashed line, wake). Both step up at $k = 1$ and are ordered by how much of the compass the encoding keeps; points are mean $\pm$ s.e.m., $n = 6$. (**c**) Place-field repetition across two observationally identical compartments ($n = 8$ each): under translation the fields repeat and the room barely decodes (0.515 against a chance of 0.5; the residual is small but reliable), under rotation the fields do not repeat and the room decodes perfectly; within-room position (R^2) stays high in both. Dots are individual networks.

less oracle, whereas a biological compass drifts and is re-anchored by vision. Two tests ask whether the fold depends on that idealisation, and it does not.

First, noise. We retrained the C_2 networks with the reported heading corrupted on 30% of steps (rotated by a random $\pm 90°$ before encoding). The dissociation is preserved: orbit-phase decoding is 0.556 under *axis* and 0.922 under *parity*, a separation of 0.367 that remains complete (Mann–Whitney $U = 0$, $p = 0.0022$ at $n = 6$ vs. 6). Relative to the clean networks the separation narrows only slightly (from 0.403), driven by a small drop in *parity* resolution ($0.955 \to 0.922$) while the folded *axis* code is unchanged ($0.552 \to 0.556$; Fig. 9**The quotient survives a degraded and a learned compass.** (**a**) Compass corruption *during training*: orbit-phase accuracy in the C_2 arena for networks that *developed* with a heading signal randomised on a given fraction of steps (chance 0.5, dashed). The non-invariant encodings (*full*, *parity*) degrade smoothly, while the invariant ones (*axis*, *const*) stay pinned at chance. This is a network that grew up with an unreliable compass, and it is *not* the lesion: an adult lesion is modelled by *silencing* the compass at test time (Fig. 11**An in-silico head-direction lesion separates what the compass *tells* the map from what it *does* to it.** Networks trained with an intact compass, which is then silenced at test time on a fraction of steps (zeroed, not randomised: an ablated head-direction system carries no signal, it does not carry a false one). $n = 6$ networks for C_1 and C_2, $n = 4$ for C_4; shading, 95% bootstrap CI over networks. (**a**) *Equivariance loss.* Orbit-phase decoding (which of two symmetry-related places is the animal in?) collapses to chance in C_2 and C_4 and holds at 0.862 in C_1, where no such pair exists ($t = 70.2$, $p = 3 \times 10^{-19}$). (**b**) *Information loss.* Orbit variance (is the quotient map still there at all?) falls by about a third in every arena. The drops are not statistically identical ($F = 10.78$, $p = 0.002$) but they are not ordered by symmetry: C_1 (-37.6%) and C_4 (-38.1%) sit at opposite ends of the symmetry range and lose indistinguishable amounts, while C_2 (-30.9%) loses least. Panels (a) and (b) together are the argument: the same lesion degrades the quotient map without regard to symmetry and folds it only where a symmetry exists. (**c**) Harland's 2×2, in silico. The *parallel* arm is the reachable null: a compass is translation-invariant by construction, so it was never holding those rooms apart, and it does not move ($0.996 \to 0.996$). The *radial* arm rises. Within-room R^2 (blue) is the positive control: it must stay positive, or the repetition number is two destroyed maps correlating with each other. It does, up to dose 0.6, and we make no claim beyond that. (**d**) Field count after a full lesion, one dot per network: fields *fall* by 11% where nothing can fold and *rise* by 16% and 20% where something can, the same dissociation Harland and Calton et al. (22) report in two different worlds.figure.caption.16), because randomising a compass an adult network already trusts contradicts its egocentric view rather than falling silent, and drives it off its training manifold. (**b**) A learned, angular-velocity compass, by arena: it resolves C_1 (0.97) but folds C_2 and C_4 to chance, building the quotient with no imposed encoding.figure.caption.14a). A compass corrupted on a third of steps folds the same symmetries a clean one does.

Second, and more strongly, we removed the oracle entirely. Rather than degrade a supplied compass, we withheld absolute heading and gave the network only *angular velocity*, the per-step turn, so it had to integrate its own heading, unanchored, as a path-integrating head-direction system does. An integrated compass is defined only up to a global rotation, so it is invariant under any rotation of the arena by construction and cannot, by itself, separate a position from its symmetry image. The learned compass reproduces the fold from first principles. In the asymmetric arena it decodes orbit phase almost perfectly (0.971, $n = 10$), because the arena's unique landmarks anchor absolute position. In the C_2 arena it falls to chance (0.526, $n = 10$): raw position is not recoverable ($R^2 = -0.08$) while position *within* the fundamental domain is ($R^2 = 0.89$), so the network knows where it is inside a copy, not which copy. In the C_4 arena it folds likewise (0.523, $n = 8$; $R^2_{\text{domain}} = -0.47$, the fourfold signature a C_2 decoder cannot see; Fig. 9**The quotient survives a degraded and a learned compass.** (**a**) Compass corruption *during training*: orbit-phase accuracy in the C_2 arena for networks that *developed* with a heading signal randomised on a given fraction of steps (chance 0.5, dashed). The non-invariant encodings (*full*, *parity*) degrade smoothly, while the invariant ones (*axis*, *const*) stay pinned at chance. This is a network that grew up with an unreliable compass, and it is *not* the lesion: an adult lesion is modelled by *silencing* the compass at test time (Fig. 11**An in-silico head-direction lesion separates what the compass *tells* the map from what it *does* to it.** Networks trained with an intact compass, which is then silenced at test time on a fraction of steps (zeroed, not randomised: an ablated head-direction system carries no signal, it does not carry a false one). $n = 6$ networks for C_1 and C_2, $n = 4$ for C_4; shading, 95% bootstrap CI over networks. (**a**) *Equivariance loss.* Orbit-phase decoding (which of two symmetry-related places is the animal in?) collapses to chance in C_2 and C_4 and holds at 0.862 in C_1, where no such pair exists ($t = 70.2$, $p = 3 \times 10^{-19}$). (**b**) *Information loss.* Orbit variance (is the quotient map still there at all?) falls by about a third in every arena. The drops are not statistically identical ($F = 10.78$, $p = 0.002$) but they are not ordered by symmetry: C_1 (-37.6%) and C_4 (-38.1%) sit at opposite ends of the symmetry range and lose indistinguishable amounts, while C_2 (-30.9%) loses least. Panels (a) and (b) together are the argument: the same lesion degrades the quotient map without regard to symmetry and folds it only where a symmetry exists. (**c**) Harland's 2×2, in silico. The *parallel* arm is the reachable null: a compass is translation-invariant by construction, so it was never holding those rooms apart, and it does not move ($0.996 \to 0.996$). The *radial* arm rises. Within-room R^2 (blue) is the positive control: it must stay positive, or the repetition number is two destroyed maps correlating with each other. It does, up to dose 0.6, and we make no claim beyond that. (**d**) Field count after a full lesion, one dot per network: fields *fall* by 11% where nothing can fold and *rise* by 16% and 20% where something can, the same dissociation Harland and Calton et al. (22) report in two different worlds.figure.caption.16), because randomising a compass an adult network already trusts contradicts its egocentric view rather than falling silent, and drives it off its training manifold. (**b**) A learned, angular-velocity compass, by arena: it resolves C_1 (0.97) but folds C_2 and C_4 to chance, building the quotient with no imposed encoding.figure.caption.14b). The fold is therefore not an artefact of imposing an invariant encoding: it is the generic consequence of a self-motion signal that carries no absolute reference under the arena's symmetry, the situation of a biological compass that has lost its landmark anchor. The hand-designed encodings are a controlled way to dial that invariance; the learned compass shows the invariance is where a real self-motion signal already lives.

Two causes leave two fingerprints, and no cell type carries the fold. The claim that folding is driven by equivariance and not information has, so far, rested on one readout. It can be made with a second that moves the *other* way, which is a stronger argument than either alone. Decompose each unit's variance into what a purely *additive* model of place and heading explains and what the full *conjunctive* model explains; the difference is nonlinear mixed selectivity, the part of the code that is a joint function of place and heading and of neither separately. It behaves nothing like the fold. Mixed selectivity is ordered by how much the compass *knows*: it rises as the compass is impoverished (0.235 under *full* at two bits, 0.292 under *parity*, 0.331 under *axis* at one bit each, 0.333 under *const* at none), and the fraction of units carrying it goes from 91% to 97%, while position-only variance falls in step ($0.505 \to 0.312$). Heading that the compass does not supply must be recovered from the conjunction of where the agent is and what it sees, and a code forced to do that mixes its variables. The rise is almost indifferent to symmetry: $+0.085$ in the C_1 arena against $+0.096$ in C_2, so 88% of it is present where nothing can fold.

Set the two readouts side by side and they come apart completely. Under *axis* and *parity*, the same one bit, phase decoding reads 0.552 and 0.955 (a difference of 0.40), while mixed selectivity reads 0.331 and 0.292 (a difference of 0.04). Across the three arenas at fixed encoding, phase decoding moves from 0.971 to 0.552 while mixed selectivity barely stirs (0.310, 0.331, 0.325). *Phase decoding is blind to information and tracks equivariance; mixed selectivity is blind to equivariance and tracks information.* They are two fingerprints of two causes, in the same networks. The consequence for reading real data is worth stating plainly, because it is otherwise got backwards: a folded code *is* more mixed than an unfolded one, but not because it folded. The two travel together only because a compass that fails to break a symmetry usually also carries less information, and it is the missing information, not the surviving symmetry, that forces the mixing.

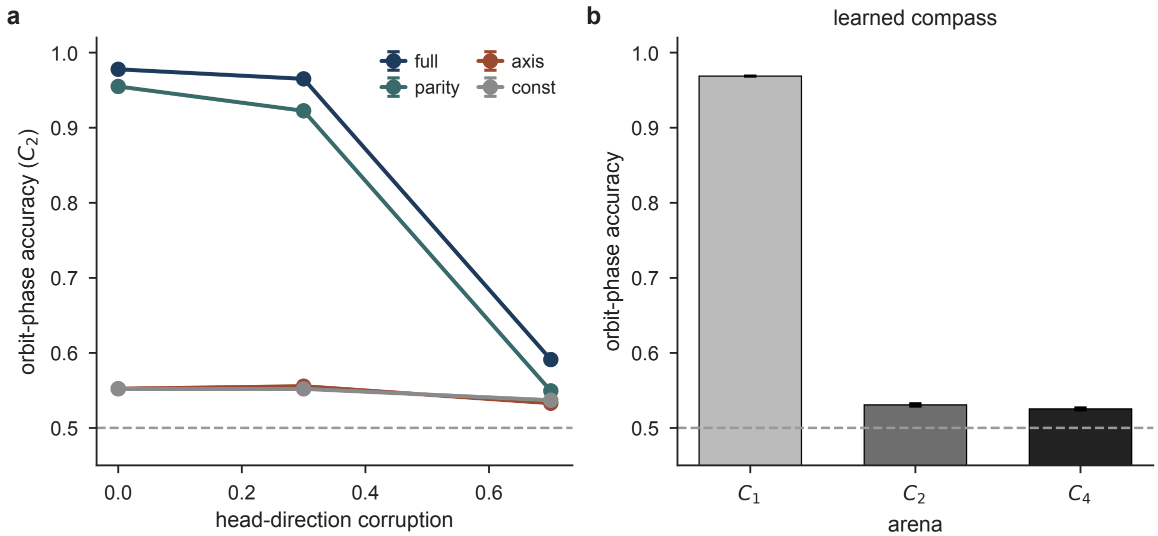


Fig. 9. The quotient survives a degraded and a learned compass. (**a**) Compass corruption *during training*: orbit-phase accuracy in the C_2 arena for networks that *developed* with a heading signal randomised on a given fraction of steps (chance 0.5, dashed). The non-invariant encodings (*full*, *parity*) degrade smoothly, while the invariant ones (*axis*, *const*) stay pinned at chance. This is a network that grew up with an unreliable compass, and it is *not* the lesion: an adult lesion is modelled by *silencing* the compass at test time (Fig. 11**An in-silico head-direction lesion separates what the compass *tells* the map from what it *does* to it.** Networks trained with an intact compass, which is then silenced at test time on a fraction of steps (zeroed, not randomised: an ablated head-direction system carries no signal, it does not carry a false one). $n = 6$ networks for C_1 and C_2, $n = 4$ for C_4; shading, 95% bootstrap CI over networks. (**a**) *Equivariance loss.* Orbit-phase decoding (which of two symmetry-related places is the animal in?) collapses to chance in C_2 and C_4 and holds at 0.862 in C_1, where no such pair exists ($t = 70.2$, $p = 3 \times 10^{-19}$). (**b**) *Information loss.* Orbit variance (is the quotient map still there at all?) falls by about a third in every arena. The drops are not statistically identical ($F = 10.78$, $p = 0.002$) but they are not ordered by symmetry: C_1 (-37.6%) and C_4 (-38.1%) sit at opposite ends of the symmetry range and lose indistinguishable amounts, while C_2 (-30.9%) loses least. Panels (a) and (b) together are the argument: the same lesion degrades the quotient map without regard to symmetry and folds it only where a symmetry exists. (**c**) Harland's 2×2, in silico. The *parallel* arm is the reachable null: a compass is translation-invariant by construction, so it was never holding those rooms apart, and it does not move ($0.996 \to 0.996$). The *radial* arm rises. Within-room R^2 (blue) is the positive control: it must stay positive, or the repetition number is two destroyed maps correlating with each other. It does, up to dose 0.6, and we make no claim beyond that. (**d**) Field count after a full lesion, one dot per network: fields *fall* by 11% where nothing can fold and *rise* by 16% and 20% where something can, the same dissociation Harland and Calton et al. (22) report in two different worlds.figure.caption.16), because randomising a compass an adult network already trusts contradicts its egocentric view rather than falling silent, and drives it off its training manifold. (**b**) A learned, angular-velocity compass, by arena: it resolves C_1 (0.97) but folds C_2 and C_4 to chance, building the quotient with no imposed encoding.

The same lesson, that an invariant compass makes the code look symmetric whether or not the arena is, disqualifies three further single-cell readouts we tried (rotational autocorrelation, isotypic odd power, and spatial representational similarity): each reports a fold in the C_1 arena, where none can exist. We relegate them and the reasoning to the Supplementary Information, and report orbit-phase decoding, whose null does survive the C_1 arena, as the primary readout throughout.

Finally, if repetition were simply a cell type seen twice, the fold should live in a subpopulation. It does not (Fig. 10**Quantifying the fold.** (**a**) Rate-map C_2 symmetry index (correlation of a unit's map with its $180°$ rotation). It saturates near 0.97 when the encoding is invariant *and* the arena is symmetric, up to the perfect-symmetry line at 1; but note the C_1 baseline, where an invariant encoding alone drives it to 0.61 though nothing can fold, so this is a descriptive readout, not a fold-specific one. (**b**) Mean place fields per unit; rises with both arena symmetry and encoding invariance, but also in the C_1 arena where nothing folds, so field count alone is confounded. (**c**) Spatial representational similarity falls as the encoding is ablated and as arena symmetry increases. Like field count, it is confounded: most of the fall is present in the C_1 arena, where nothing folds, so it evidences that the code stays spatial rather than that it folds. (**d**) Cross-seed correlation between independently trained networks stays above 0.96 throughout (y zoomed to 0.9–1.0): descriptive evidence that the fold is reproducible rather than degenerate. No null is reported for this measure and none is claimed. Arenas $C_1/C_2/C_4$; bars are mean $\pm$ s.e.m. over networks ($n = 6$ per encoding for C_1 and C_2, $n = 4$ for C_4; 64 networks in all). The unit-level symmetry-index distributions reported in the text are computed per unit over the full 112-network sweep (56,000 units).figure.caption.15). Scored unit by unit as the correlation of a rate map with its own $180°$ rotation, across all 56,000 units the fold is everywhere: in the C_2 arena under *axis*, 93.1% of units score above 0.8 (median 0.94, fifth percentile 0.77), while under *full*, the reachable null, 54.3% of units fall *below* 0.2 (median 0.09), so the readout is free to return the negative answer. And the fold does not track cell type: the most boundary-tuned quartile of units folds at 0.947 and the least at 0.923, a difference of 0.024 on an effect of size 0.9, and the correlation between border tuning and folding is $r = 0.16$ (under 3% of the variance), no larger than its correlations with spatial information (0.25) or nonlinear mixing (-0.15). No cell class carries the fold, because it is a property of what the code can distinguish, not of any unit's receptive field.

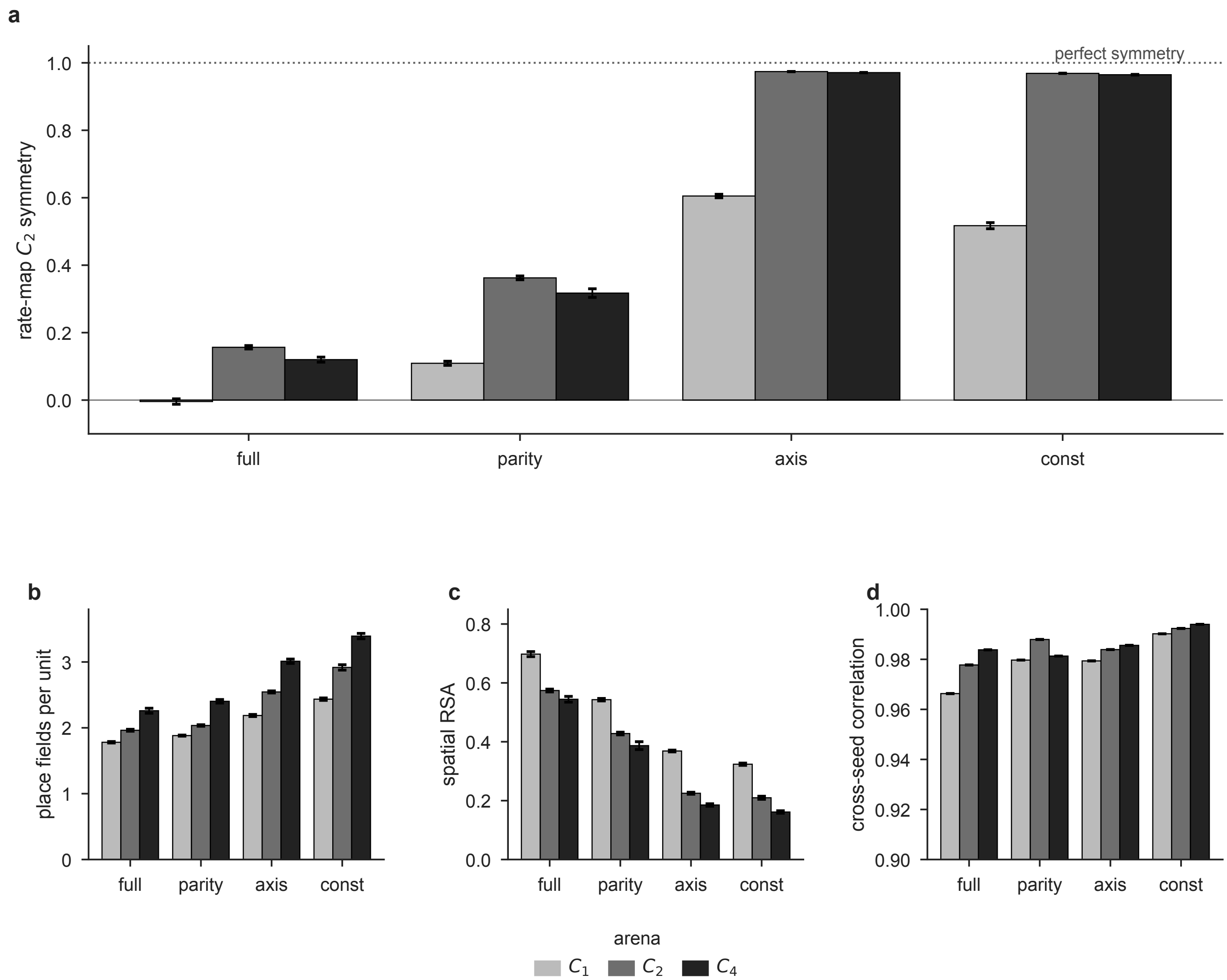
Cast in the standard hippocampal metrics, the same population reads as a realistic map that fails to remap exactly where the compass is blind. The population-vector correlation (15) between a position and its symmetry image is 0.98 under the invariant encodings in the C_2 and C_4 arenas and about 0.28 under *full*, with *parity* intermediate (0.58 in C_2; Supplementary Information, Fig. S13). Fields per unit rise from 1.78 under *full* in C_1 to 3.39 under *const* in C_4 (Fig. 10**Quantifying the fold.** (**a**) Rate-map C_2 symmetry index (correlation of a unit's map with its $180°$ rotation). It saturates near 0.97 when the encoding is invariant *and* the arena is symmetric, up to the perfect-symmetry line at 1; but note the C_1 baseline, where an invariant encoding alone drives it to 0.61 though nothing can fold, so this is a descriptive readout, not a fold-specific one. (**b**) Mean place fields per unit; rises with both arena symmetry and encoding invariance, but also in the C_1 arena where nothing folds, so field count alone is confounded. (**c**) Spatial representational similarity falls as the encoding is ablated and as arena symmetry increases. Like field count, it is confounded: most of the fall is present in the C_1 arena, where nothing folds, so it evidences that the code stays spatial rather than that it folds. (**d**) Cross-seed correlation between independently trained networks stays above 0.96 throughout (y zoomed to 0.9–1.0): descriptive evidence that the fold is reproducible rather than degenerate. No null is reported for this measure and none is claimed. Arenas $C_1/C_2/C_4$; bars are mean $\pm$ s.e.m. over networks ($n = 6$ per encoding for C_1 and C_2, $n = 4$ for C_4; 64 networks in all). The unit-level symmetry-index distributions reported in the text are computed per unit over the full 112-network sweep (56,000 units).figure.caption.15b); the field count in the unfolded baseline is under-dispersed (Fano factor 0.65, decisively non-Poisson), a respect in which the model departs from the closer-to-Poisson statistics reported for large environments (16). Scored by the criteria applied to recordings, 76–96% of units qualify as place cells and about 30% are border-modulated, stable across encodings and arenas, and independently trained networks agree above 0.96 in every condition (Fig. 10**Quantifying the fold.** (**a**) Rate-map C_2 symmetry index (correlation of a unit's map with its $180°$ rotation). It saturates near 0.97 when the encoding is invariant *and* the arena is symmetric, up to the perfect-symmetry line at 1; but note the C_1 baseline, where an invariant encoding alone drives it to 0.61 though nothing can fold, so this is a descriptive readout, not a fold-specific one. (**b**) Mean place fields per unit; rises with both arena symmetry and encoding invariance, but also in the C_1 arena where nothing folds, so field count alone is confounded. (**c**) Spatial representational similarity falls as the encoding is ablated and as arena symmetry increases. Like field count, it is confounded: most of the fall is present in the C_1 arena, where nothing folds, so it evidences that the code stays spatial rather than that it folds. (**d**) Cross-seed correlation between independently trained networks stays above 0.96 throughout (y zoomed to 0.9–1.0): descriptive evidence that the fold is reproducible rather than degenerate. No null is reported for this measure and none is claimed. Arenas $C_1/C_2/C_4$; bars are mean $\pm$ s.e.m. over networks ($n = 6$ per encoding for C_1 and C_2, $n = 4$ for C_4; 64 networks in all). The unit-level symmetry-index distributions reported in the text are computed per unit over the full 112-network sweep (56,000 units).figure.caption.15d; descriptive, no null is claimed). The multi-field, spatially repeating cells the fold produces are the in-silico counterpart of the repeated place fields recorded in symmetric and multicompartment environments (3, 17).

Discussion

The learned cognitive map is a quotient. When the environment has a symmetry the agent's self-motion cannot break, the predictive code writes each orbit of symmetry-related positions as a single place, and what it maps is X/G rather than X. This is a statement about group actions, not information channels: G is fixed by the self-motion signal, which resolves exactly the symmetries under which it is not itself invariant. A compass is blind to translation, since moving the world does not change a heading, and so cannot separate translationally repeated compartments; it is not blind to rotation, and can separate rotationally related ones. The principle is not about head direction. Any self-motion signal resolves exactly the symmetries of the environment under which it is not invariant: a pure speed cue resolves almost nothing, a place-modulated efference copy would fold no arena. Which symmetries a learned map collapses is a property of the agent's sensorimotor access to space, not of the hippocampus; head direction is simply the access we could vary cleanly.

What the principle explains in the animal data. The dissociation the theory predicts is already in the record, in two studies. Spiers et al. (2) recorded CA1 place cells in four visually identical compartments in a row and found "a high degree of spatial repetition with a slight degree of rate-based discrimination"; those compartments are translationally related, and the authors conclude "path integration is not used, or at least not spontaneously" to separate them. Grieves et al. (3) then ran the same four compartments in two arrangements. Parallel, translationally related: "place cells often exhibited repeated fields." Radial, at $60°$ and so rotationally related: "significantly less place field repetition was apparent" (all $p < 0.0001$). The dissociation survived removal of the orientation landmark, so it is not carried by extra-maze vision, and the authors attribute it to "directional information derived from the animal's self-motion." Our account supplies the reason: a compass is invariant under translation and equivariant under rotation, so the input process is G-equivariant in the parallel arrangement and not in the radial one, and a predictive code must fold in the first case and may lift in the second. Repetition is not a failure of the hippocampus; it is what a predictive code must do when its self-motion signal cannot see the symmetry.

The same study reports a behavioural consequence of the fold, not only a neural one. In a separate cohort, rats learned an odour–location discrimination that required telling the four compartments apart, and learning "was significantly impaired" in the parallel arrangement relative to the radial one: fewer animals reached criterion, and those that did needed many more sessions. Grieves et al. (3) conclude that "place field repetition constrains spatial learning." A folded code carries a behavioural cost, tested directly, in the same paper, with the same manipulation. Two things follow that the animal data alone cannot establish: the effect is a property of *invariance*, not of how much the compass tells you (two



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encodings carrying the same bit dissociate completely, Table 2Orbit-phase decoding accuracy (chance $= 0.500$). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.table.caption.5), and it is a property of *prediction* (under reconstruction the compass is inert, Table 5Orbit-phase decoding in the C_2 arena as a function of prediction horizon k (chance $= 0.500$, $n = 6$ per cell).table.caption.12).

The dissociation can be produced without touching the animal, by changing only which group element relates the compartments, and that experiment exists too. Fuhs et al. (18) ran rats between two identical boxes in a curtained enclosure, counterbalanced within animals on the same days: side by side (a translation), or abutted after rotating each by 90° in opposite senses so their orientations differed by 180° (a rotation). The rats *walked* between the boxes in both, so self-motion was available throughout and only the group element changed. Under translation there was no remapping: the median between-box correlation was 0.71, identical to the within-box 0.71 ($p > 0.5$), present already in the first two days. Under rotation the maps separated completely (median between-box correlation 0.06, indistinguishable from complete remapping). Fuhs and colleagues draw the conclusion the account needs, in their own terms: "linear path integration appears easily overridden by visual landmarks, whereas angular path integration is more sensitive to cue discordance." Self-motion does not break all symmetries equally: heading is a strong breaker of rotations, displacement a weak breaker of translations. That asymmetry is not something we impose on the theory but the empirical finding this experiment was built to make. Two honest qualifications. The remapping was *delayed* in two of three rats, appearing only on the second or third visit, and the authors are explicit that it "cannot be explained as a direct consequence of a sensory manipulation"; our networks model the asymptotic state, not the transition to it, and before remapping the rotated box carried a 180°-rotated *copy* of the first box's map, an anchoring phenomenon our account can name but our model does not produce. And Skaggs and McNaughton (19) report partial remapping in the same-orientation geometry, which Fuhs et al. failed to replicate; a weak breaker is precisely what should give a graded and unreliable effect.

The causal test has already been run. The account yields a lesion prediction, and it is a *signed and selective* one rather than a main effect. A compass is invariant under translation by construction and equivariant under rotation, so abolishing it should do opposite things in the two arrangements. In the parallel arrangement the compass was never breaking the symmetry, so removing it should change nothing. In the radial arrangement the compass is the only thing breaking the symmetry, so removing it should make rotation join translation as a symmetry the animal cannot resolve, and the fold should appear where intact animals show almost none.

That experiment exists. Harland et al. (10) made bilateral lesions of the lateral mammillary nuclei, which abolish the head-direction signal rather than merely degrading it: the anterodorsal tuning curves go flat and never recover (20), and no cell in a lesioned animal retains direction-modulated firing (21). In the parallel compartments the lesion did nothing: 65% of compartment pairs correlated at $r \geq 0.5$ in shams against 63% in lesioned animals ($t(10) = 1.06$, $p = 0.31$). In the radial compartments it restored the fold: 82% of sham correlations were low ($r \leq 0.5$), whereas 44% of lesioned cells correlated at $r \geq 0.5$ ($t(5.2) = 3.30$, $p = 0.021$). The maze-by-group interaction is large ($F(1,10) = 13.60$, $p < 0.005$, $\eta^2 = 0.58$), and the effect sizes against the shuffle move exactly as the account requires: parallel repetition is unchanged by the lesion ($d = 1.08$ sham, 1.02 lesioned) while radial repetition rises from nothing to intermediate ($d = 0.07$ sham, 0.53 lesioned). Two features make this a test of invariance rather than of similarity: the radial rate maps were rotated into alignment before being correlated, so the measure asks directly whether the code is invariant under the group element relating the compartments, and the apparatus stood in a curtained enclosure with no polarising cue, so no landmark was available to break the symmetry in the animal's place. The prediction, its null half included, is what the data show, with no parameter fitted.

A real lesion, though, removes information and equivariance in one stroke, and cannot say which does the folding. In the model they can be held apart, and the account needs them to do two separable things: losing the compass's *information* should degrade the map *everywhere*, symmetry or not, while losing its *equivariance* should fold the map *only where a symmetry exists*. We measured this by silencing the compass of networks that developed with it intact. Silencing, not corrupting: an ablated head-direction system carries no signal (21), and because our observations are egocentric, a randomised compass contradicts the view and drives the network off its manifold (we report that condition as a control; it produces a spurious fold, since two destroyed maps correlate). The two consequences dissociate (Fig. 11**An in-silico head-direction lesion separates what the compass *tells* the map from what it *does* to it.** Networks trained with an intact compass, which is then silenced at test time on a fraction of steps (zeroed, not randomised: an ablated head-direction system carries no signal, it does not carry a false one). $n = 6$ networks for C_1 and C_2, $n = 4$ for C_4; shading, 95% bootstrap CI over networks. (**a**) *Equivariance loss.* Orbit-phase decoding (which of two symmetry-related places is the animal in?) collapses to chance in C_2 and C_4 and holds at 0.862 in C_1, where no such pair exists ($t = 70.2$, $p = 3 \times 10^{-19}$). (**b**) *Information loss.* Orbit variance (is the quotient map still there at all?) falls by about a third in every arena. The drops are not statistically identical ($F = 10.78$, $p = 0.002$) but they are not ordered by symmetry: C_1 (-37.6%) and C_4 (-38.1%) sit at opposite ends of the symmetry range and lose indistinguishable amounts, while C_2 (-30.9%) loses least. Panels (a) and (b) together are the argument: the same lesion degrades the quotient map without regard to symmetry and folds it only where a symmetry exists. (**c**) Harland's 2×2, in silico. The *parallel* arm is the reachable null: a compass is translation-invariant by construction, so it was never holding those rooms apart, and it does not move ($0.996 \to 0.996$). The *radial* arm rises. Within-room R^2 (blue) is the positive control: it must stay positive, or the repetition number is two destroyed maps correlating with each other. It does, up to dose 0.6, and we make no claim beyond that. (**d**) Field count after a full lesion, one dot per network: fields *fall* by 11% where nothing can fold and *rise* by 16% and 20% where something can, the same dissociation Harland and Calton et al. (22) report in two different worlds.figure.caption.16; $n = 6$ networks each for C_1 and C_2, $n = 4$ for C_4). Orbit variance (is the quotient map still there?) falls by about a third in every arena: -37.6% in C_1, -30.9% in C_2, -38.1% in C_4. These are not identical ($F = 10.78$, $p = 0.002$), but crucially they are not *ordered by symmetry*: the two arenas at opposite ends of the range lose indistinguishable amounts (-37.6% vs -38.1%, $U = 15$, $p = 0.61$), while the one between loses least (C_2 vs C_4, $U = 24$, $p = 0.010$). Phase accuracy behaves the opposite way and is ordered by symmetry exactly: it falls from 0.986 to 0.566 in C_2 and 0.985 to 0.567 in C_4, while in C_1 it holds at 0.862 ($t = 70.2$, $p = 3 \times 10^{-19}$). Field count follows the same split, falling by 11% where nothing can fold and rising by 16% and 20% where something can. The information loss therefore cannot be what folds the map: it is comparable in an arena with no symmetry and one with a fourfold symmetry, and the fold happens in only one of them. Harland's intact rats say the same thing from the other side: they ran in a rotationally symmetric radial layout, exactly where an invariant channel would fold without any surgery, and their place fields did *not* repeat. The map was reading the compass that breaks the symmetry, and only when that compass is gone does it fall back on the one that does not.

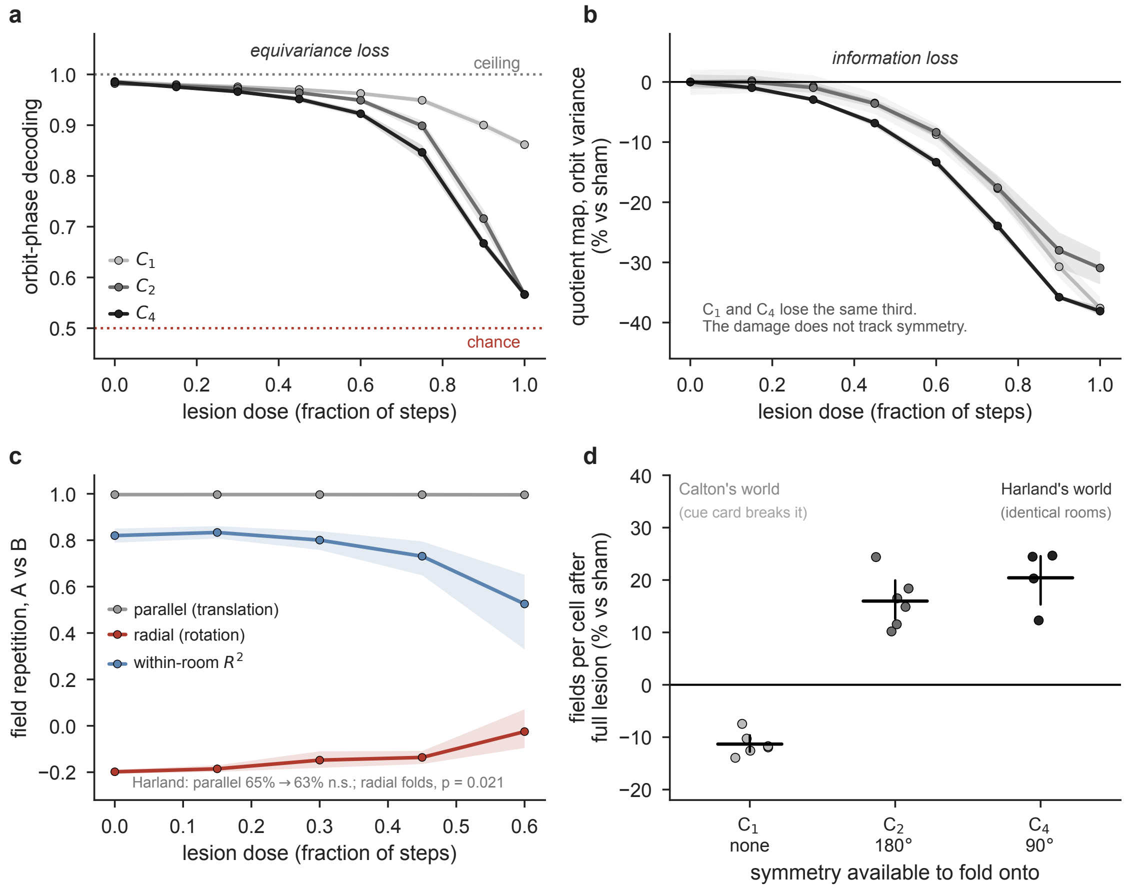

Fig. 11. An in-silico head-direction lesion separates what the compass *tells* the map from what it *does* to it. Networks trained with an intact compass, which is then silenced at test time on a fraction of steps (zeroed, not randomised: an ablated head-direction system carries no signal, it does not carry a false one). $n = 6$ networks for C_1 and C_2, $n = 4$ for C_4; shading, 95% bootstrap CI over networks. (**a**) *Equivariance loss.* Orbit-phase decoding (which of two symmetry-related places is the animal in?) collapses to chance in C_2 and C_4 and holds at 0.862 in C_1, where no such pair exists ($t = 70.2$, $p = 3 \times 10^{-19}$). (**b**) *Information loss.* Orbit variance (is the quotient map still there at all?) falls by about a third in every arena. The drops are not statistically identical ($F = 10.78$, $p = 0.002$) but they are not ordered by symmetry: C_1 (-37.6%) and C_4 (-38.1%) sit at opposite ends of the symmetry range and lose indistinguishable amounts, while C_2 (-30.9%) loses least. Panels (a) and (b) together are the argument: the same lesion degrades the quotient map without regard to symmetry and folds it only where a symmetry exists. (**c**) Harland's 2×2, in silico. The *parallel* arm is the reachable null: a compass is translation-invariant by construction, so it was never holding those rooms apart, and it does not move ($0.996 \to 0.996$). The *radial* arm rises. Within-room R^2 (blue) is the positive control: it must stay positive, or the repetition number is two destroyed maps correlating with each other. It does, up to dose 0.6, and we make no claim beyond that. (**d**) Field count after a full lesion, one dot per network: fields *fall* by 11% where nothing can fold and *rise* by 16% and 20% where something can, the same dissociation Harland and Calton et al. (22) report in two different worlds.

One principle, two lesion studies that seem to disagree. The account also resolves a discrepancy the lesion literature leaves standing. Harland's lesioned place cells carried less spatial information (1.29 to 1.05 bits per spike) and were less sparse (0.32 to 0.37), and their fields repeated. Yet anterodorsal and postsubicular lesions leave place-field number (1.29 to 1.53), size, rate and sparsity statistically unchanged, and produce no repetition at all (22). Read as claims about what a head-direction lesion does, these disagree. They stop disagreeing once one asks what world each lesion was performed in. Calton et al. recorded in a cylinder whose inside was "featureless, except for a white card occupying $\sim 100°$ of arc", a polarising landmark at a definite bearing, which breaks the rotational symmetry. Harland's compartments were identical, behind a black curtain, with no polarising cue: the symmetry stands. A compass lesion cannot create a symmetry; it can only stop breaking one already there. Where some other channel has already broken the symmetry, destroying the compass degrades the map and folds nothing; where the compass was the only thing breaking it, destroying it folds the map. The two studies are the same lesion in two different worlds, and one principle predicts the null and the positive. The account also predicts an experience-dependent *unfolding*: Carpenter et al. (23) found that the grid code across two connected compartments begins repeated and becomes a single global representation with experience. In our terms the repeated code is the quotient, which unfolds once the animal acquires any cue (a faint distal landmark, a learned context) that breaks the symmetry the compass could not. The fold is a starting point set by self-motion, not a fixed endpoint.

The invariant compass is real, and it is not the one the map reads. The central manipulation of this paper is a compass that carries a full bit of heading yet is *invariant* under the symmetry (*axis*), set against one carrying the same bit and breaking it (*parity*). This invites the objection that no such thing exists, that a G-invariant compass is a device built to make a theorem land. It exists. Zhang et al. (24) recorded retrosplenial cells whose own tuning mirrored the layout's symmetry: bidirectional (two peaks 180° apart) in a twofold two-box ($48/575$ cells) and tetradirectional (four peaks at 90°) in a fourfold four-box ($67/743$). Three qualifications decide the weight this bears, and we state all three. The comparison across the two symmetric layouts is between cells, not within them. The comparison that *is* within cell is against a onefold arena, and there the pattern nearly disappears: of the cells multidirectional in a symmetric box, only $1/8$ stayed bidirectional and $3/65$ tetradirectional above shuffle. And the symmetry the cells track is the visual-geometric one: the authors broke the global symmetry with odour (lemon in one compartment pair, vanilla in the other) and the cells stayed fourfold regardless. The pattern is learned rather than immediately sensory: absent in naive animals, appearing only after experience of the layout, and largely surviving darkness ($21/28$ cells in the first dark trial). A directional signal *approximately* invariant under a symmetry of the environment exists in mammalian cortex; we say approximately and mean it, and every claim built on it inherits that. LaChance and Hasselmo (25) produce the same object more cleanly, by changing the symmetry and holding the cells fixed: duplicating a cue card so two identical cards face each other across a square makes postrhinal head-direction cells fire in two opposite directions ($n = 34$, $W = 1$, $p = 2 \times 10^{-10}$) and retrosplenial ones likewise ($n = 37$, $W = 111$, $p = 1 \times 10^{-4}$), while entorhinal head-direction cells show no detectable increase ($n = 30$, $W = 207$, $p = 0.62$, a failure to reject, on a test with no reported power, which we treat as such). The second peak is reliably weaker than the original ($p = 4 \times 10^{-5}$ postrhinal, $p = 7 \times 10^{-11}$ retrosplenial), so the cortical compass is approximately, not exactly, invariant.

This same literature supplies the strongest objection to the account: the invariant compass does not replace the ordinary one, it *coexists* with it. Classic unidirectional head-direction cells are recorded in the same retrosplenial tissue, electrophysiologically distinct (peak-to-trough $163\,\mu$s against $402\,\mu$s), and they never fold. Zhang et al. (24) draw the consequence themselves: "the preservation of a global HD signal also allows for the possibility of disambiguation." If an unfolded compass survives, X is available downstream, and the fold should never happen. The objection is answered by taking the theory literally. The claim is not that *the* compass folds, but that a predictive code represents X/G where G is the subgroup the self-motion signal *it reads* cannot break. Two directional populations, one invariant and one breaking, are not an embarrassment; they are the theory's two conditions realised side by side, and they turn the matter into an empirical question: which does the hippocampal map read? The lesion answers it. Destroying the head-direction system removes the *breaking* population; if the map had been reading the invariant channel, the lesion would change nothing, and instead it folds the map (10). (We flag that nobody has lesioned the head-direction system and recorded the invariant cells, so their survival through the lesion is an assumption of this argument, not a measurement.) Anatomy points the same way but must be stated narrowly. The multidirectional cells are overwhelmingly a *dysgranular* population (76%; 24), and the retrosplenial output characterised by anterograde tracing terminates in the retrohippocampal region (post-, pre- and parasubiculum and entorhinal cortex) rather than the subiculum that builds the place map (26, 27). What does *not* follow is that dysgranular cortex fails to reach the hippocampus at all: Li et al. (28) report that its one projection to CA1 is *GABAergic*, with no reciprocal excitatory route, and Sugar et al. (29) decline to read a missing tracing entry as a proven absence. The defensible claim is the narrow one, that there is no substantial direct *excitatory* route from the cells carrying the invariant code to the cells building the map, and even it is about Zhang's population only; postrhinal cortex, where LaChance and Hasselmo's fold is strongest, *does* project monosynaptically to the subiculum (30), and there the work is done by the lesion, not the anatomy.

This forces a correction to how Zhang et al. (24) should be read, including by us. Their multidirectional cells are not the *antecedent* of the law, the invariant compass whose reading folds the map. They are its *directional shadow*: a directional code whose tuning has taken on the symmetry of the group its own input channel cannot resolve. The law applies to them, not through them. The distinction is not pedantry, and our networks enforce it in both directions. A directional tuning curve is *not sufficient* to infer a map fold: give a network the *axis* compass in the C_1 arena, with no symmetry at all, and its units are already 93% bidirectional while the map does not fold (phase 0.971); put the same compass in the C_2 arena and the tuning barely moves (0.997) while the map collapses (0.552): the same tuning curve on opposite maps. Nor is it *necessary*, the half we nearly missed: the most completely folded networks, *const* in C_4, sit on the $1/4$ ceiling with *flat* tuning ($m_1 = 0.011$, $m_2 = 0.009$), because a C_4-symmetric function of four headings is a constant. A maximally folded map can present with no directional tuning, and a completely unfolded one as strongly bidirectional. The consequence is a warning that cuts against every account, ours included: *no multidirectional tuning curve in cortex licenses an inference about the hippocampal map*, and neither does its absence. It is also what dissolves the apparent paradox in LaChance and Hasselmo (25), where the postrhinal compass folds hard while entorhinal grid maps a synapse away do not move. There is no paradox, because a folded compass upstream says nothing about a map that reads a different one. It is why the lesion, a fact about the map, is the load-bearing evidence here. We must be equally clear about a mechanism we do *not* reproduce and cannot rule out: Alexander et al. (31) propose that an egocentric boundary-vector cell, sampled by an animal that hugs walls, looks fourfold tuned in a square from directional occupancy alone, with no invariant compass. Our networks have boundary cells but not the sampling bias. Our agent samples headings near uniformly, and removing the occupancy correction moves tuning barely at all (*const*/C_1 0.230 raw against 0.203 corrected; *axis*/C_1

0.895 against 0.929), so their account is untested here rather than refuted; settling it needs a thigmotactic agent we have not built. What we can say is that their mechanism, whatever it explains about *tuning*, has no account of Harland's lesion, which is a fact about the *map*.

Relation to other models of the spatial code. The parallel-versus-radial dissociation discriminates among models, because the two arrangements are built from the same compartments and are locally identical. Any model that fixes firing from local sensory or boundary geometry must predict the same repetition in both. The sharpest such objection is that we have rediscovered the boundary-vector-cell account of repetition (32) with extra steps; a predictive network does grow boundary-vector cells (33), and ours are abundant (41.6% of units in C_1 under *full* are better fit by a boundary-vector cell than any place field, rising to 61.2% in C_4). But a boundary-vector cell is tuned to a wall at an *allocentric* bearing, and an allocentric bearing is undefined without a compass; as Julian et al. (34) put it, such models "assume the existence of a reliable head-direction signal." Our networks show exactly that dependence: ablating the compass drops the boundary-cell fraction from 41.6% to 14.0% in C_1 and 61.2% to 0.8% in C_4 (`bvc_tuning.csv`), so the boundary code is built on the compass, not underneath it. The decisive observation is the matched contrast (Fig. 12**Boundary-vector cells are downstream of the compass.** (**a**) Fraction of units better fitted by a single idealised boundary-vector tuning curve (a Gaussian over the distance to the wall at a given allocentric bearing) than by any single Gaussian place field; the two families have the same free parameters. One dot per network. The network grows boundary cells, and they require the compass: the fraction falls from 41.6% to 14.0% in the C_1 arena and from 61.2% to 0.8% in the C_4 arena as the compass is ablated. (**b**) The decisive contrast. Boundary-cell fraction against orbit-phase accuracy, with the group-theoretic floor and the non-invariant ceiling both drawn. *axis* and *parity* carry the same one bit and support comparable boundary populations, yet only *axis* folds. Same boundary code, opposite maps: a boundary-vector model has no representation of *which* bit the compass carries, and so cannot produce this dissociation.figure.caption.17): *axis* and *parity* support comparable boundary populations (29.2% and 22.1% in C_2, *parity* if anything the better fit, $r = 0.482$ against 0.449), yet *axis* folds and *parity* does not. Same boundary code, opposite maps, which no boundary-vector model can produce, because it has no representation of *which* bit the compass carries. The recent local-feature version of the objection (25), that a square has four walls so a wall detector fires four times, with no compass or symmetry group needed, is answered on the same ground and two of theirs. The symmetric cortical code does not reach the structures that would inherit it (the entorhinal head-direction null, $W = 207$, $p = 0.62$, is load-bearing; the grid null is weaker because a hexagonal lattice is inversion-symmetric). The local-feature and quotient accounts make identical predictions in a square, and the discriminating shape, one where local features and global symmetry disagree, the authors themselves name as necessary future work but did not run. And our own *axis*/*parity* dissociation sees the same walls and folds oppositely.

Boundary-vector cells are therefore the mechanism a predictive objective discovers; the quotient law says what any such mechanism can and cannot tell apart. Read the other way, the result *instantiates* the successor representation rather than competing with it. The successor representation (5, 35) is predictive, but over a state space that is supplied rather than inferred from aliased observations, so whether two symmetric compartments are one state or two is an assumption, not a result. Under G-equivariance, with a G-invariant prior and a policy measurable with respect to the aliased belief, the agent's belief process descends to a Markov process on X/G, because G-related states are belief-indistinguishable. This is the symmetry-induced Markov decision process homomorphism of reinforcement learning (8, 36, 37), and the equivalence it induces, in which states are collapsed exactly because they agree on the entire future, is a bisimulation (38), the coarsest abstraction preserving an MDP's dynamics (9), computed here not by an explicit bisimulation metric but as the by-product of a network trained only to predict. That the few-percent residual above the floor is a convergence gap and not a leaky fold is something the input bound settles: Table 4The bound of Eq. 1equation.0.1, computed from the arenas with no free parameters, against the trained networks. Every encoding falls to the accuracy its input-distinguishability predicts; the mean absolute error across the twelve cells is 0.031. (We report the bound as a bound, not a fit: the predicted values are near-degenerate, clustered at 1.0 and at the folding floors, so a correlation coefficient across cells would merely register that separation.)table.caption.11's distinguishable fraction is exactly 0.000 for *axis* and *const* in C_2, so the observations are *exactly* bisimilar and the residual can only be the network's finite-training approximation of the exact quotient, sitting near the quotient ceiling rather than the un-quotiented one (Table 2Orbit-phase decoding accuracy (chance = 0.500). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.table.caption.5). Two learned-map models place against the same test. The clone-structured cognitive graph (39, 40) disambiguates aliased observations by transition context and reproduces repetition, but not the dissociation: run without a compass, Raju et al. (40) report that "place field repetition persists in the modified layout," folding even when the rooms are rotated. They then supply the fix themselves, that "an external head direction input" makes the different orientation give unique place fields. That is our claim, reached independently in a model with no boundaries and no metric. The Tolman–Eichenbaum machine (41) factorises a structural code that is a function of the action sequence, and an action code invariant under the arena's symmetry folds it as ours folds, so the two agree on the effect and differ only in mechanism. The symmetry itself has been noticed before: Jeffery (42) observes that repetition between same-orientation compartments "is a translational symmetry, which echoes the translational symmetry of the environment," while "the head direction signal has a rotational asymmetry." That is the intuition this paper formalises. What is new is not the obser-

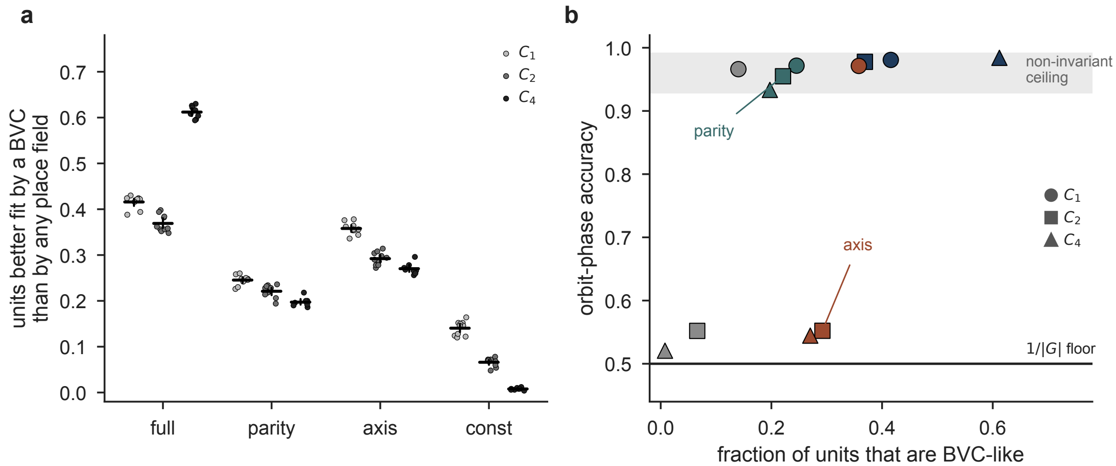

Fig. 12. Boundary-vector cells are downstream of the compass. (**a**) Fraction of units better fitted by a single idealised boundary-vector tuning curve (a Gaussian over the distance to the wall at a given allocentric bearing) than by any single Gaussian place field; the two families have the same free parameters. One dot per network. The network grows boundary cells, and they require the compass: the fraction falls from 41.6% to 14.0% in the C_1 arena and from 61.2% to 0.8% in the C_4 arena as the compass is ablated. (**b**) The decisive contrast. Boundary-cell fraction against orbit-phase accuracy, with the group-theoretic floor and the non-invariant ceiling both drawn. *axis* and *parity* carry the same one bit and support comparable boundary populations, yet only *axis* folds. Same boundary code, opposite maps: a boundary-vector model has no representation of *which* bit the compass carries, and so cannot produce this dissociation.

vation but the derivation: that the relevant object is the subgroup of the environment's symmetry the self-motion signal fails to break, that a learned predictive code must collapse onto its quotient, that the collapse is bounded at $1/|G|$, and that what governs it is equivariance rather than information, so two encodings carrying the same single bit come apart completely (Table 6Where models of the spatial code stand on the symmetry dissociation. The decisive column is the last: the parallel and radial arrangements are locally identical, so any model that reads firing from local geometry or the current observation predicts the same repetition in both, and cannot produce the dissociation that Grieves et al. (3) observed.table.caption.18).

What the model reproduces, and what it does not. Table 7The model against the animal literature. "Reproduced" means the model shows the phenomenon without being fitted to it; "absent" means the phenomenon cannot exist in this architecture and we do not claim it; "tested, not found" means the architecture could in principle show the phenomenon but a direct test did not find it.table.caption.19 scores the model against the animal literature. Most rows are reproduced without fitting, but three kinds of departure are worth naming. The two rows marked *absent* are architectural: head direction enters as an oracle input rather than being learned, and no non-negativity constraint is imposed, which is the condition under which hexagonal grids emerge in trained path integrators (43). Nothing here bears on the ring or toroidal manifolds of entorhinal cortex, and we make no claim about them.

The row marked *tested, not found* is the one place the model and the animals part on the fold itself. Spiers reported repetition *with* a slight rate-based discrimination; our network, given perfectly identical rooms, keeps none, and folds to within a small residual. Because real compartments are never perfectly identical, we reasoned a cognitive map should collapse toward X/G, not exactly onto it, and tested whether a symmetry-breaking channel too weak to lift position would still modulate rate, the signature Spiers reported. Eq. 1equation.0.1 lets that curve be predicted before training. A first attempt (a uniform brightness shift) gave a step rather than a ramp, so we replaced it with fixed per-pixel Gaussian noise competing against a room-B tint swept to target single-step discriminabilities evenly over $d' \in \{0, \ldots, 3\}$ (Methods). Both a generalisation and a within-trajectory room decoder then rise smoothly with d' (Spearman $\rho = 0.92$ and 0.99, $n = 64$, $p < 10^{-25}$; $d' = 0$ vs $d' = 3$, $p = 0.000155$), so the sweep tests the prediction as designed. The prediction is not borne out. Field-shape repetition stays at ceiling across the sweep (≈ 0.997, $p = 0.88$, $\rho = 0.17$), and a rate-asymmetry readout is flat throughout ($p = 0.88$, $\rho = 0.02$): no evidence of a rate-based signal, let alone one that leads position. This is an honest negative on the rate-leads-position prediction, as pre-registered. Reconciliation with Spiers' partial fold has to come, in this architecture, from the residual already present at zero added cue (generalisation room decode 0.510, above chance, Wilcoxon $p = 0.023$, $n = 8$), not from a graded rate-versus-position dissociation: some trace of compartment identity survives even the perfectly-identical limit, but it does not grow into a rate code as the cue strengthens.

Limitations. A first group of limitations are architectural absences on the same footing as the missing grid torus. The arenas are discrete 18×18 grids with four headings, so the head-direction "ring" has four points and no continuous topology; nothing here speaks to the toroidal manifolds of entorhinal cortex (44), and the place-like but not grid-like tuning (full-compass gridness: mean -0.29, maximum $+0.08$, none above 0.3; invariant encodings can reach 0.50 through symmetry-induced field repetition rather than hexag-

Table 6. Where models of the spatial code stand on the symmetry dissociation. The decisive column is the last: the parallel and radial arrangements are locally identical, so any model that reads firing from local geometry or the current observation predicts the same repetition in both, and cannot produce the dissociation that Grieves et al. (3) observed.

model class	fields learned from experience	repetition in identical rooms	parallel-vs-radial dissociation
Continuous attractor / path integration (6, 43)	metric built in	not addressed	no
Boundary-vector / geometry (7)	partly	yes	no (predicts repetition in both)
Successor representation (5)	yes (predictive)	depends on the given state space	not addressed
Clone-structured graph (39, 40)	yes (transition clones)	yes (aliased views share clones)	no: repeats in *both* without a compass (40)
Tolman–Eichenbaum machine (41)	yes (structural)	yes, if the action code is symmetry-invariant	consistent
Reconstruction / autoencoder (this work, control)	yes	yes	no (predicted; folds every encoding alike in the rotational arena, $p = 0.59$; not run on the compartment mazes)
Predictive RNN (1), this work	yes (emergent)	yes	**yes**, set by HD invariance

onal tuning) is expected given a supplied compass and no non-negativity constraint (43). The network has no membrane dynamics, spikes, theta, or excitatory–inhibitory separation, and is trained once on a fixed arena, so it speaks to the representation a predictive objective converges to, not to drift across sessions; all results use one architecture (hidden size 500, horizon $k = 5$), and hidden and arena size are not swept.

A second group are empirical departures from biology, most already bounded by a control. The supplied compass is tested by the 30%-corruption and learned-compass results; what is not modelled is a continuous compass that drifts and is re-anchored over time. Real open-field place cells are largely omnidirectional (47, 48); this model's units are only partly so (opposite-heading rate-map correlation, *full*, C_1: mean 0.56, median 0.64, 35% of pairs above 0.8), because head direction enters as a direct input rather than being merged away upstream. A proposed topology-before-geometry developmental order does not hold here under a shuffle null (position is linearly decodable at $R^2 = 0.79$ before training), so we make no such claim, and place fields do not anticipate future position (Table 7The model against the animal literature. "Reproduced" means the model shows the phenomenon without being fitted to it; "absent" means the phenomenon cannot exist in this architecture and we do not claim it; "tested, not found" means the architecture could in principle show the phenomenon but a direct test did not find it.table.caption.19).

A third group bound how far the results generalise. Symmetry here is a property of the landmark layout, not the arena geometry, which is square in every condition, so only the C_1/C_2/C_4 arenas satisfy the Theorem's hypotheses exactly; the compartment mazes are an empirical demonstration of the same principle rather than a strict instance, since the connecting corridor lacks the group's cyclic closure (a temporal-conditioning analysis rules out corridor memory as the alternative, Methods). A sharper control asks whether room decodability on the compartment maze rises with horizon: for $k \in \{0, 1, 3\}$ ($n = 8$ each) steady-state room decodability stays at chance (0.508, 0.502, 0.510; Spearman $\rho = 0.07$, $p = 0.73$) while repetition holds near ceiling (0.98–0.99), supporting architecture-robustness over horizon-limitation, though $k = 5$ and $k = 10$ remain untested. And the one real-neuron test here, the CA1 city-block reanalysis, is exploratory rather than diagnostic: a conjunctive place-by-direction code predicts the same correlation, two controls confirm the accounts are not separable in these data, and we treat the model's own prediction, not the reanalysis, as the evidence.

Outlook. The appeal of predictive learning is that a single domain-general objective, predict the next observation, grows structured spatial representations with no hand-built spatial prior, in artificial networks as in the hippocampus (1, 5). What this study adds is that the geometry of the learned code is then set by the symmetry structure of the agent's interaction with the world, the group action of the environment on the self-motion signal, rather than by the architecture. That makes the paradigm a controlled developmental probe: because the representation is learned, each factor that shapes it in an animal, and is there confounded or fixed, becomes a variable that can be turned on its own, whether the geometry of the enclosure, the information and invariance of the self-motion signal, the locomotor policy, or the training task.

Two experiments follow immediately, both within reach of a laboratory already equipped for them. The first is a prediction we would stake the account on, and it is sharper than it looks because the law makes *two different* predictions in one apparatus. LaChance and Hasselmo (25) showed that two identical cue cards on opposite walls make postrhinal and retrosplenial head-direction cells bidirectional, a cortical compass invariant under a 180° rotation. The tempting move is to predict a fold and go record place cells there. That would be wrong, and the account says why: in the same animals, entorhinal head-direction cells did *not* become bidirectional ($W = 207$, $p = 0.62$), so the classical compass, carried forward by path integration, still breaks the symmetry, and the classical compass is the one the map reads. In an *intact* rat the two-cue square should therefore produce no fold, but *anchoring*: with the animal disoriented, the map must choose between the two 180°-related gauges the scene allows, and should adopt one at chance, which is what Keinath et al. (49) report. The fold appears only when the breaking compass is removed. The experiment is the crossing of two apparatuses that already exist: $\{$one cue, two cues$\} \times \{$sham, head-direction lesion$\}$. Repetition is predicted in exactly one of the four cells (two cues, lesioned), with nothing free to fit. Two are already observed: the intact single-cue square is an ordinary place map, and the lesioned single-cue square is Calton et al. (22), who found no multiplication of fields because a cue-controlled arena has no symmetry to fold onto. Two assumptions do work here and we would rather name them: that the hippocampus is dominated by the classical breaking compass rather than the folded postrhinal one (the lesion shows the map reads *a* breaking compass, not *which*, and postrhinal cortex reaches the subiculum monosynaptically (30)), and that the entorhinal null holds in a session entered from a single-cue baseline, whereas the anchoring prediction requires between-session disorientation, so the design must specify which it is. The cortical fold also shows hysteresis (25), so the law is a claim about a *learned* map, and the conditions must be counterbalanced rather than run in sequence. The fourth cell, two cues and lesioned, is the one genuinely open, and it is the one the account stands on; a null there would be a serious problem for us, and we say so. Neither LaChance and Hasselmo (25) nor Zhang et al. (24) recorded a place cell, and the laboratories holding the lesion and the place cells (10) are not the ones holding the cue-duplication apparatus. The experiment is a collaboration away, not a decade.

The second is a control the model can run and the field has not. In Zhang et al. (24) the local shape of each compartment is confounded with the global symmetry of the layout: the fourfold environment is built of *square* compartments and the twofold one of *rectangles*, so a code tracking local walls and a code tracking global symmetry predict the same thing. The confound is severable in simulation, and only there without years of rat training: build a fourfold layout out of compartments whose own shape has no fourfold symmetry, and ask whether the compass and the map follow the global group or the local geometry. The same severance answers LaChance and Hasselmo (25), whose authors name such a shape as necessary future work but did not run one. We have not run it here, and it is the first thing we will run next. Progressively more biological designs (a learned compass allowed to drift, non-negative rates, separate path-integration and sensory streams, planning modules) would then let one ask which ingredients are necessary for the joint emergence of place and grid cells, turning the catalogue of spatial cell types into consequences that follow from an objective, an architecture, and an experience, of which the symmetry-driven fold reported here is one.

Methods

Architecture and training. We use `pRNN_th` (1) unchanged: a layer-normalised recurrent cell, hidden size 500, truncation $T = 200$, rollout horizon $k = 5$ unless stated, observation size 147 ($7 \times 7 \times 3$), action size 5. Training uses RMSprop ($\alpha = 0.95$, $\epsilon = 10^{-6}$) with four parameter groups at per-group learning rates $\lambda\sqrt{1/H}$ for $W_{\text{rec}}, W_{\text{out}}$, $\lambda\sqrt{1/(n_{\text{obs}} + n_{\text{act}})}$ for W_{in} and 0.1λ for the bias, with bias-only weight decay; $\lambda = 2 \times 10^{-3}$, batch size 8, gradient clipped per model to norm 1, 80,000 steps.

Arenas. 18×18 MiniGrid arenas (324 positions) with landmark layouts of C_1, C_2 and C_4 rotational symmetry. The observational discriminability index (ODI; the Spearman correlation between heading-averaged observation distance and spatial distance across all pairs of positions) is *not* matched across the three arenas, and cannot be: symmetric landmarks make distinct positions observationally identical, so discriminability necessarily falls with symmetry order. Measured over all 324 positions, ODI $= 0.336$ (C_1), 0.288 (C_2), 0.262 (C_4).
This does not confound the results, for two reasons. The decisive contrast, *axis* vs. *parity*, is made *within* one arena, on the same trajectories, so the observation stream and its ODI are identical between the two arms; the only difference is a 4×4 matrix on the heading block. And across arenas, ODI does not predict the readout: with a full compass, phase decoding is 0.981, 0.978 and 0.984 for C_1, C_2, C_4 (Pearson $r = -0.37$ against ODI, $p = 0.76$, $n = 3$ arenas), with the least discriminable arena scoring highest. Trajectories: 10,000 per arena, 200 steps, random start position and heading. The policy is a fixed-probability random walk (forward 0.60, turn left 0.15, turn right 0.15, stop 0.10), identical at every position, heading, and arena; nothing about it depends on which copy of the fundamental domain the agent occupies, which is what makes it measurable with respect to the agent's belief (Theorem).

Head-direction encodings. The heading block of the action vector is left-multiplied by a fixed 4×4 matrix: identity (*full*); $\frac{1}{2}(I + P^2)$ where P is the cyclic shift (*axis*); the block-averaging matrix pairing $\{E, S\}$ and $\{W, N\}$ (*parity*); and $\frac{1}{4}\mathbf{1}\mathbf{1}^\top$ (*const*). Column sums are preserved, so activation magnitude is unchanged. Entropies under a uniform heading prior are $2, 1, 1, 0$ bits.

Ensemble training. All networks sharing a computation graph (arena, encoding and seed all leave it unchanged; the horizon k does not) are trained as a single job using `torch.func.stack_module_state` with `vmap(functional_call)`, which lowers every per-model matrix multiply to one batched multiply and makes the kernel-launch count independent of the number of models. Forward passes are bit-exact against separate runs; gradients are bit-exact with noise disabled and agree to 2.3×10^{-10} absolute with the production noise on (`bmm` reassociation). Per-model gradient clipping, the four-group RMSprop and the sum-of-per-model-means loss are all preserved; each is checked against the reference implementation by test.

Ensemble sizes differ across analyses, and each figure and table states its own n: the orbit-phase and quotient-geometry results use the full sweep ($n = 10$ per encoding in C_1 and C_2, $n = 8$ in C_4; 112 networks in all), whereas the C_4 four-way ceiling (Table 3C_4 four-way phase decoding in the s4 arena (chance $= 0.250$); $n = 4$ networks per encoding.table.caption.7) and the per-network fold and lesion analyses (Fig. 10**Quantifying the fold.** (**a**) Rate-map C_2 symmetry index (correlation of a unit's map with its 180° rotation). It saturates near 0.97 when the encoding is invariant *and* the arena is symmetric, up to the perfect-symmetry line at 1; but note the C_1 baseline, where an invariant encoding alone drives it to 0.61 though nothing can fold, so this is a descriptive readout, not a fold-specific one. (**b**) Mean place fields per unit; rises with both arena symmetry and encoding invariance, but also in the C_1 arena where nothing folds, so field count alone is confounded. (**c**) Spatial representational similarity falls as the encoding is ablated and as arena symmetry increases. Like field count, it is confounded: most of the fall is present in the C_1 arena, where nothing folds, so it evidences that the code stays spatial rather than that it folds. (**d**) Cross-seed correlation between independently trained networks stays above 0.96 throughout (y zoomed to 0.9–1.0): descriptive evidence that the fold is reproducible rather than degenerate. No null is reported for this measure and none is claimed. Arenas $C_1/C_2/C_4$; bars are mean $\pm$ s.e.m. over networks ($n = 6$ per encoding for C_1 and C_2, $n = 4$ for C_4; 64 networks in all). The unit-level symmetry-index distributions reported in the text are computed per unit over the full 112-network sweep (56,000 units).figure.caption.15, Fig. 11**An in-silico head-direction lesion separates what the compass *tells* the map from what it *does* to it.** Networks trained with an intact compass, which is then silenced at test time on a fraction of steps (zeroed, not randomised: an ablated head-direction system carries no signal, it does not carry a false one). $n = 6$ networks for C_1 and C_2, $n = 4$ for C_4; shading, 95% bootstrap CI over networks. (**a**) *Equivariance loss.* Orbit-phase decoding (which of two symmetry-related places is the animal in?) collapses to chance in C_2 and C_4 and holds at 0.862 in C_1, where no such pair exists ($t = 70.2$, $p = 3 \times 10^{-19}$). (**b**) *Information loss.* Orbit variance (is the quotient map still there at all?) falls by about a third in every arena. The drops are not statistically identical ($F = 10.78$, $p = 0.002$) but they are not ordered by symmetry: C_1 (-37.6%) and C_4 (-38.1%) sit at opposite ends of the symmetry range and lose indistinguishable amounts, while C_2 (-30.9%) loses least. Panels (a) and (b) together are the argument: the same lesion degrades the quotient map without regard to symmetry and folds it only where a symmetry exists. (**c**) Harland's 2×2, in silico. The *parallel* arm is the reachable null: a compass is translation-invariant by construction, so it was never holding those rooms apart, and it does not move ($0.996 \to 0.996$). The *radial* arm rises. Within-room R^2 (blue) is the positive control: it must stay positive, or the repetition number is two destroyed maps correlating with each other. It does, up to dose 0.6, and we make no claim beyond that. (**d**) Field count after a full lesion, one dot per network: fields *fall* by 11% where nothing can fold and *rise* by 16% and 20% where something can, the same dissociation Harland and Calton et al. (22) report in two different worlds.figure.caption.16; $n = 6$ in C_1 and C_2, $n = 4$ in C_4) use smaller matched subsets.

Orbit-phase decoding. Hidden states are collected over 20,000 raw timesteps (no per-position averaging); repeated visits to one position therefore contribute multiple, slightly different rows, and `GroupKFold` groups every row from an orbit's two positions together regardless of how many visits each contributed. Orbits are $\{x, R^2 x\}$ under the 180° rotation (or the four-element C_4 orbits where stated); the arena side is even, so no rotation has an integer fixed point and every orbit has exactly $|G|$ cells (162 orbits at C_2, 81 at C_4). Phase labels are balanced, and cross-validation folds hold out whole orbits, so the decoder must generalise a phase direction rather than memorise orbits or exploit a repeated visit to the same position appearing on both sides of a split. We report the accuracy of a logistic decoder, together with the R^2 of ridge decoders for the raw position and for the position within the fundamental domain. The latter is not a positive control (recovering a folded coordinate from an unfolded code requires a nonlinear map), so the sanity check that a code still represents space is $\max(R^2_{\text{raw}}, R^2_{\text{domain}})$.
We checked directly whether this pipeline is itself biased, since a 500-dimensional feature decoded by 5-fold `GroupKFold` over only 162 orbit groups is a regime where an improperly validated decoder could read above chance on a truly invariant code. The standardiser is fit on the training fold only (never on the pooled data before the split), and no other preprocessing step (no PCA, no whitening, no cross-validated regularisation strength) touches held-out rows before scoring, so there is no pre-split leakage in the code. We also ran the decoder on a synthetic code matched to this exact regime, 500 dimensions, 162 C_2 orbits, 20,000 rows, constructed to be exactly invariant to phase by hand: it reads 0.498, indistinguishable from the nominal 0.500. The decoder is therefore unbiased in the regime this paper uses it in. The k=0 full-compass condition's 0.536 and every folded network's 0.52–0.56 (Table 2Orbit-phase decoding accuracy (chance $= 0.500$). $n = 10$ per cell for s1 and s2, $n = 8$ for s4.table.caption.5) are not, consequently, a decoder artefact to be read against as an empirical floor; they are real signal in the trained networks themselves, a small residual above the nominal chance the permutation test above confirms is not noise. We attribute it, as in the Discussion, to finite training never reaching the exactly invariant solution a population-level, infinitely-trained network would.

Within-episode phase (Experiment 0). The across-episode analysis above shows absolute orbit phase is undecodable in steady state, but that alone cannot distinguish a code that never carries phase information from one that carries it locally, right after an episode begins, and loses it as the trajectory accumulates. We tested this directly with a per-episode-relative label, $z(t) = \text{phase}(t) \oplus \text{phase}(0)$, decoded with a logistic model under `GroupKFold`-by-episode and `class_weight='balanced'`, reporting balanced accuracy (chance $= 0.5$ regardless of z's base rate) as a function of dwell time since episode start. An earlier version of this analysis, reporting raw accuracy, was invalidated by a majority-class confound: under a random walk, $P(z=1)$ rises from about 0.08 near episode start to about 0.46 by $100+$ steps purely from position continuity, independent of the hidden state, so a majority-class-only classifier scored about 0.93 at 0–10 steps using zero real information; comparing every reported number against this baseline is now standard in the pipeline. The corrected result, $n = 10$ seeds each of *learned*, *axis* and *const* in the C_2 arena, is nominally above chance (Wilcoxon signed-rank against 0.5, exact two-sided, uncorrected) at every dwell bin tested for all three encodings, decaying from 0.66–0.74 near episode start toward the across-episode baseline (0.52–0.53) by $100+$ steps. The effect at the longest bin is markedly weaker for *learned* ($p = 0.049$, at the edge of detection) than for *axis*/*const* ($p \leq 0.0098$), though none actually falls to non-significance. The fold is therefore decodable-but-drifting: the network keeps a locally coherent orbit-frame for a window after an episode starts, which attenuates toward the steady-state floor rather than vanishing at the first step (an instantaneous fold) or never appearing at all (a code with no phase information to lose).

Isotypic spectrum. Each unit's rate map is decomposed into the four characters of C_4; P_k is the normalised power in character k, $\sum_k P_k = 1$, and $P_1 = P_3$ for real maps. $\mathrm{RA} = P_0 - P_2$ and $\mathrm{odd} = P_1 + P_3$.

Compartment mazes. Room-interior observational identity (two and four rooms) is verified by direct enumeration: rendering every room-interior cell and heading and comparing pixel images across rooms gives zero mismatch (0/144 pairs, two rooms; 0/432, four rooms). The connecting corridor is not: because every room's corridor opens onto one shared junction, a room at either end of the junction's span sees it truncated on one side while an interior room does not, and once the agent's forward view (F cells) reaches far enough into the corridor to include the junction, this asymmetry becomes visible (maximum single-channel pixel mismatch of 5/255, first appearing exactly where the view depth reaches the junction column). Room-interior views therefore satisfy the aliasing this maze is built to test; the maze as a whole, corridor included, does not instantiate a finite group acting freely on the full state space the way the rotationally symmetric C_2/C_4 arenas do, so the two- and four-room results are an empirical demonstration of the same principle rather than a direct instance of the Theorem. Room decoding (`label_rooms`) is restricted to room-interior timesteps for exactly this reason.

The two-room arrangements compound this: the corridor connecting the compartments is not the same length in both. The translation arrangement's doors are joined by a direct link (27 cells, door to door); the rotation arrangement's link runs around three edges of the arena to avoid cutting through the rotated room's wall ring (57 cells; 58 non-room cells in total against 28 for translation). A longer, more distinctive corridor is a channel by which recurrent memory of the just-completed transit, not the compass, could carry room identity into the following room-interior states, which would confound the translation/rotation contrast with corridor length rather than isolating head-direction equivariance. We tested this directly on the trained networks by conditioning room decoding on the number of steps since the animal last crossed into its current room. The two arrangements show opposite temporal signatures. Under translation, decoding is above chance immediately on entry (0.56 within the first two steps) and decays to the near-chance floor within about ten steps (0.51, matching the steady-state value reported in the main text); a corridor-memory trace is real but small and transient. Under rotation, decoding is *below* its ceiling immediately on entry (0.71) and *rises* to it over the following several steps (0.92, 0.98, 0.99 at 2–5, 5–10 and $10+$ steps; `corridor_dwell.csv`). A corridor-memory account predicts the opposite ordering, highest immediately after the longer transit and decaying, so it does not explain the rotation result; the rise instead matches a network that needs a few steps of position-and-heading integration after entry to resolve which room it is in, consistent with the compass account this maze is built to test. The corridor-length asymmetry is therefore a real, small, and transient confound specific to the translation arrangement, not an alternative explanation for the dissociation itself: it makes the reported translation number (averaged over all room-interior timesteps, including the contaminated first few) a slightly conservative estimate of the steady-state fold, and it does not touch the rotation result at all.

CA1 reanalysis. From Hockeimer et al. (11) (JHU repository doi:10.7281/T15HMQD4, `AlleySuperpopDirVisitFiltered.csv`) we took, for each place field, its per-visit firing rates split by travel direction, and formed a directional index $\mathrm{DI} = (r_+ - r_-)/(r_+ + r_-)$ with $+/-$ the two directions of the field's alley (E/W for horizontal alleys, N/S for vertical); fields with fewer than three visits in either direction were dropped. The primary statistic is the Pearson correlation over within-cell same-orientation field-DI *pairs* (a cell's repeating fields lie in translation-related alleys). The shuffle null permutes DI values across all repeating fields *within an orientation*, breaking the assignment of DIs to cells while preserving the marginal DI distribution, and is recomputed 2000 times. This null, however, also destroys cell-level directionality, so we ran two further controls that hold it fixed. First, the same correlation on *non-repeating* cells' same-orientation fields (which the fold does not predict should correlate). Second, a directionality-matched scramble that preserves each cell's per-orientation mean DI and resamples the within-cell residuals from the global pool, isolating the between-cell (global-directionality) contribution; the directional index is substantially a between-cell property (mixed-effects ICC 0.31), and this scramble reproduces (indeed exceeds) the observed correlation. The observed correlation is therefore consistent with the quotient but not separable from a conjunctive place-by-direction code in these data; we report it as such.

Statistics. All contrasts are two-sided exact Mann–Whitney U tests on per-network values. The planned contrasts are *axis* vs. *parity* within each arena, at $n = 10$ per cell for s1 and s2 and $n = 8$ for s4; at $n = 10$ vs. 10 the exact two-sided floor is $2/\binom{20}{10} = 1.1 \times 10^{-5}$, and a p-value at that floor indicates complete separation (every network in one group above every network in the other). Bonferroni correction across the three planned arena contrasts is applied where stated. Secondary experiments carry their own n: the prediction-horizon and noisy-compass sweeps use $n = 6$ per cell (floor 0.00216), the isotypic re-run $n = 5$ (floor 0.0079). Folded decoding accuracies are compared to their $1/|G|$ floor by a one-sample t-test where a residual is claimed. No results were excluded; models failing a criterion are reported as such rather than dropped.

Data and code availability

The source data underlying every figure and table are provided as comma-separated files in `project5_symmetry/Report/data/`; a per-file index maps each reported value to its source (`Report/data/README.md`). All code, the analysis pipeline that regenerates those files from trained checkpoints, and the figure-generation scripts are openly available at https://github.com/RavulapalliManas/Thesis_work (`project5_symmetry/`), and will be archived on Software Heritage on acceptance. The test suite pins the claims that fail silently: the bit-exactness of the batched ensemble against per-model training, that the $k = 0$ target is exactly the anchor observation, that a synthetic C_2-folded code reads chance phase while a code with no spatial signal is distinguishable from it, and that the isotypic identity $\mathrm{RA} = P_0 - P_2$ holds. Trained checkpoints and raw trajectories ($\sim$40 GB) are available from the author on request; every published number is reproducible from the committed data files without them.

Author contributions

M.V.S.R. designed the study, implemented the models and analyses, ran the experiments and wrote the manuscript.

Competing interests

The author declares no competing interests.

Use of AI tools

Large language models were used for code review, refactoring, and drafting assistance. All experiments, statistics and scientific claims were specified, executed and verified by the author; every number reported here is produced by the archived analysis scripts.

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Table 7. The model against the animal literature. "Reproduced" means the model shows the phenomenon without being fitted to it; "absent" means the phenomenon cannot exist in this architecture and we do not claim it; "tested, not found" means the architecture could in principle show the phenomenon but a direct test did not find it.

observation in animals	source	in this model	verdict
Fields repeat across translationally related identical compartments; path integration does not separate them	Spiers et al. (2)	translation leaves the compass invariant, so the code folds onto X/G	reproduced
Repetition in parallel compartments, markedly less in radial ones; survives removal of the orientation landmark; attributed to self-motion	Grieves et al. (3)	rotation makes the compass equivariant, so the code lifts	reproduced
Place fields repeat across the arms of a hairpin maze, and the repetition is running-direction dependent	Derdikman et al. (17)	folding onto X/G makes one unit fire at all $\|G\|$ orbit-mates, and the direction dependence is the compass entering the fold, but we have not built and run a hairpin-maze arena to test this directly	consistent
Anterodorsal and postsubicular lesions leave place-field number, size, rate and sparsity unchanged, and lower coherence, in a cylinder polarised by a cue card	Calton et al. (22)	his cue card breaks the rotational symmetry, so his is our C_1 arena and there is nothing for the lesion to fold onto: silencing the compass there leaves the mean rate unchanged, lowers coherence, and crucially does *not* multiply fields, in flat contrast to the symmetric arenas. But it does not leave field count untouched either: ours *falls* by 11.3% ($p = 0.031$, $n = 6$), where his is unchanged with a point estimate that rises ($1.29 \to 1.53$, n.s.)	partly: the *absence of multiplication* is reproduced, the field count itself is not, and the sign is wrong
The same lesion *raises* in-field directional information ($0.25 \to 0.48$ bits per spike)	Calton et al. (22)	**not reproduced.** Ours falls to near zero. In our network the compass is an additive input coordinate, so a unit's directional tuning is inherited from it and must vanish with it. Calton et al. propose the opposite role, that head direction *cancels* the directionality of egocentric views and binds them into an allocentric code, so that removing it causes "directional fragmentation". Ours is a model of what the compass makes representable, not of how it binds views, and this readout is outside what it can speak to	mismatch
Two visually identical boxes, walked between, are not distinguished when related by a translation and remap completely when related by a 180° rotation	Fuhs et al. (18)	the compass is invariant under the first group element and equivariant under the second; the compartment networks repeat under translation and lift under rotation	reproduced
Head direction plus recurrence plus prediction are needed for offline replay	Levenstein et al. (1)	offline trajectories exceed the coverage of a wake path of equal length only with the full compass and $k \geq 1$; coverage falls monotonically as the compass is ablated ($1.30 \to 0.80 \to 0.45 \to 0.31\times$ wake), and the autoencoder never leaves $0.37\times$. Replay appears at the same horizon as symmetry resolution and improves no further with a longer one	reproduced
A predictive objective is sometimes read as implying prospective (anticipatory) place coding	—	rebuilding each rate map on the position up to five steps ahead does not sharpen it at any prediction horizon; the network's predictive character shows up in offline replay, not in a spatial lead of its fields	tested, not found
Rate remapping: a slight rate-based discrimination between otherwise repeated fields	Spiers et al. (2)	the pure predictive model folds almost completely (repetition $+0.997$, compartment decodable only at 0.515 against a chance of 0.5); a graded-cue sweep (Discussion) found no	tested, not found